41

Quantum Mechanics

CHAPTER OUTLINE

41.1 The Wave Function

41.2 Analysis Model: Quantum Particle Under Boundary Conditions

41.3 The Schrödinger Equation

41.4 A Particle in a Well of Finite Height

41.5 Tunneling Through a Potential Energy Barrier

41.6 Applications of Tunneling

41.7 The Simple Harmonic Oscillator

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ41.1 Answer (b). Fewer particles are reflected as the height of the potential barrier decreases and approaches the energy of the particles. By Equations 41.22 and 41.23, the transmission coefficient $T \approx e^{-2CL}$, where $C = \sqrt{2m(U-E)}/\hbar$, increases as $U - E$ decreases, so the reflection coefficient $R = 1 - T \approx 1 - e^{-2CL}$ decreases as $U - E$ decreases.

OQ41.2 The ranking is answer (b) > (a) > (c) > (e) > (d). From Equation 41.14, consider the quantity

$$E = \left(\frac{h^2}{8mL^2}\right)n^2:$$

(a) $$\left[\frac{h^2}{8m_1(3\text{ nm})^2}\right](1)^2 = \frac{1}{9}\left(\frac{h^2}{8m_1}\text{ nm}^{-1}\right)$$

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(b) $\left[\frac{h^2}{8m_1(3\text{ nm})^2}\right](2)^2 = \frac{4}{9}\left(\frac{h^2}{8m_1}\text{ nm}^{-1}\right)$

(c) $\left[\frac{h^2}{8(2m_1)(3\text{ nm})^2}\right](1)^2 = \frac{1}{18}\left(\frac{h^2}{8m_1}\text{ nm}^{-1}\right)$

(d) $\left[\frac{(0)^2}{8m_1(3\text{ nm})^2}\right](1)^2 = 0$

(e) $\left[\frac{h^2}{8m_1(6\text{ nm})^2}\right](1)^2 = \frac{1}{36}\left(\frac{h^2}{8m_1}\text{ nm}^{-1}\right)$

OQ41.3 (a) True. Examples: An electron has mass and charge, but it can also display interference effects.

(b) False. An electron has rest energy $E_R = m_e c^2$.

(c) True. A moving electron possesses kinetic energy.

(d) True. $p = m_e u$.

(e) True.

OQ41.4 (a) True. Examples: A photon behaves as a particle in the photoelectric effect and as a wave in double-slit interference.

(b) True. A photon cannot have rest energy (mass) because it is never at rest: it travels at the speed of light.

(c) True. $E = hf$.

(d) True. $p = E/c$.

(e) True.

OQ41.5 Answer (d). The probability of finding the particle is at the antinodes (places of greatest amplitude) of the standing wave.

OQ41.6 Compare the ground state wave functions in Figures 41.4 and 41.7 in the text. In the square well with infinitely high walls, the particle's simplest wave function has strict nodes separated by the length L of the well. The particle's wavelength is $2L$, its momentum $h/2L$, and its energy $p^2/2m = h^2/8mL^2$. In the well with walls of only finite height, the wave function has nonzero amplitude at the walls, and it extends outside the walls.

(i) Answer (a). The ground state wave function extends somewhat outside the walls of the finite well, so the particle's wavelength is longer.

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(ii) Answer (b). The particle's momentum in its ground state is smaller because $p = h/\lambda$ and the wave function has a larger wavelength.

(iii) Answer (b). The particle has less energy because is has smaller momentum.

OQ41.7 Answer (e). From the relation between the square of the wave function and the probability P of finding the particle in the interval Δx = (7 nm − 4 nm) = 3 nm, we have

$$|\psi|^2 \Delta x = P \qquad \rightarrow \qquad \psi = \sqrt{\frac{P}{\Delta x}} = \sqrt{\frac{0.48}{3 \text{ nm}}} = 0.40 \text{ nm}^{-1}$$

OQ41.8 Answer (a). Because of the exponential tailing of the wave function within the barrier, the tunneling current is more sensitive to the width of the barrier than to its height. Notice that the exponent term CL in the transmission coefficient $T \approx e^{-2CL}$, where $C = \sqrt{2m(U-E)}/\hbar$, decreases more if L decreases than if U decreases by the same percentage.

OQ41.9 Answer (c). Other points see a wider potential-energy barrier and carry much less tunneling current.

OQ41.10 Answer (d). The probability of finding the particle is greatest at the place of greatest amplitude of the wave function. The next most likely place is point *b*, after that, points *a* and *e* appear to be equally probable. The particle would never be found at point *c*.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ41.1 Consider the Heisenberg uncertainty principle. It implies that electrons initially moving at the same speed and accelerated by an electric field through the same distance *need not* all have the same measured speed after being accelerated. Perhaps the philosopher could have said "it is necessary for the very existence of science that the same conditions always produce the same results within the uncertainty of the measurements."

CQ41.2 Consider a particle bound to a restricted region of space. If its minimum energy were zero, then the particle could have zero momentum and zero uncertainty in its momentum. At the same time, the uncertainty in its position would not be infinite, but equal to the width of the region. In such a case, the uncertainty product $\Delta x \Delta p_x$ would be zero, violating the uncertainty principle. This contradiction proves that the minimum energy of the particle is not zero.

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CQ41.3 The motion of the quantum particle does not consist of moving through successive points. The particle has no definite position. It can sometimes be found on one side of a node and sometimes on the other side, but never at the node itself. There is no contradiction here, for the quantum particle is moving as a wave. It is not a classical particle. In particular, the particle does not speed up to infinite speed to cross the node.

CQ41.4 (a) $\psi(x)$ becomes infinite as $x \to \infty$.

(b) $\psi(x)$ is discontinuous and becomes infinite at $x = \pi/2, 3\pi/2, \ldots$

CQ41.5 A particle's wave function represents its state, containing all the information there is about its location and motion. The squared absolute value of its wave function tells where we would classically think of the particle as spending most its time. $|\Psi|^2$ is the probability distribution function for the position of the particle.

CQ41.6 In quantum mechanics, particles are treated as wave functions, not classical particles. In classical mechanics, the kinetic energy is never negative. That implies that $E \geq U$. Treating the particle as a wave, the Schrödinger equation predicts that there is a nonzero probability that a particle can tunnel through a barrier—a region in which $E < U$.

CQ41.7 Both (d) and (e) are not physically significant. Wave function (d) is not acceptable because ψ is not single-valued. Wave function (e) is not acceptable because ψ is discontinuous (as is its slope).

CQ41.8 Newton's 1st and 2nd laws are used to determine the motion of a particle of large mass. The Schrödinger equation is not used to determine the motion of a particle of small mass; rather, it is used to determine the state of the wave function of a particle of small mass. In particular, the states of atomic electrons are confined-wave states whose wave functions are solutions to the Schrödinger equation. Anything that we can know about a particle comes from its wave function.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 41.1 The Wave Function

P41.1 (a) The wave function,

$$\psi(x) = Ae^{i\left(5\times10^{10}x\right)} = A\cos\left(5\times10^{10}x\right) + iA\sin\left(5\times10^{10}x\right)$$

will go through one full cycle between $x_1 = 0$ and $(5.00 \times 10^{10})x_2 = 2\pi$. The wavelength is then

$$\lambda = x_2 - x_1 = \frac{2\pi}{5.00\times10^{10}\ \text{m}^{-1}} = \boxed{1.26\times10^{-10}\ \text{m}}$$

To say the same thing, we can inspect $Ae^{i\left(5\times10^{10}x\right)}$ to see that the wave number is $k = 5.00 \times 10^{10}\ \text{m}^{-1} = 2\pi/\lambda$.

(b) Since $\lambda = h/p$, the momentum is

$$p = \frac{h}{\lambda} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{1.26\times10^{-10}\ \text{m}} = \boxed{5.27\times10^{-24}\ \text{kg}\cdot\text{m/s}}$$

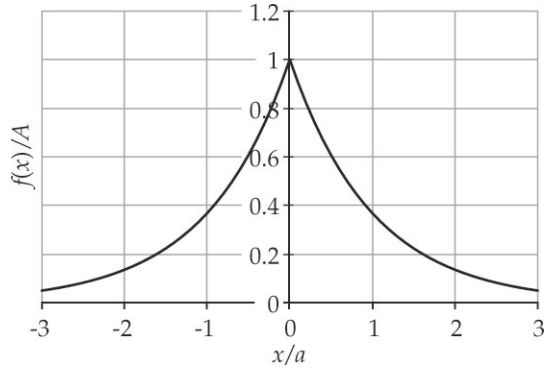
(c) The electron's kinetic energy is

$$K = \frac{1}{2}mu^2 = \frac{p^2}{2m}$$

$$= \frac{\left(5.27\times10^{-24}\ \text{kg}\cdot\text{m/s}\right)^2}{2\left(9.11\times10^{-31}\ \text{kg}\right)}\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right) = \boxed{95.3\ \text{eV}}$$

[We use u to represent the speed of a particle with mass in chapters 39, 40, and 41.]

P41.2 (a) See ANS. FIG. P41.2 for a graph of $\frac{f(x)}{A} = e^{-|x|/a}$ for the range $-3 < \frac{x}{a} < 3$.



ANS. FIG. P41.2

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(b) Normalization requires

$$\int_{\text{all space}} |\psi|^2 \, dx = 1:$$

$$\int_{-\infty}^{\infty} A^2 e^{-2|x|/a} dx = 2\int_{0}^{\infty} A^2 e^{-2|x|/a} dx = 1$$

$$-aA^2 e^{-2|x|/a}\Big|_0^{\infty} = aA^2 = 1 \quad \rightarrow \quad A = \boxed{\frac{1}{\sqrt{a}}}$$

(c) $$P = \int_{-a}^{a} \frac{e^{-2|x|/a}}{a} dx = 2\int_{0}^{a} \frac{e^{-2|x|/a}}{a} dx = -e^{-2x/a}\Big|_0^a = -e^{-2} + 1 = \boxed{0.865}$$

P41.3 (a) Normalization requires

$$\int_{\text{all space}} |\psi|^2 \, dx = 1:$$

$$\int_{0}^{1.00} A^2 x^2 dx = 1$$

$$\frac{A^2 x^3}{3}\Big|_0^{1.00} = \frac{A^{23}}{3} = 1 \qquad \rightarrow \qquad \boxed{A = \sqrt{3}}$$

(b) $$P = \int_{0.300}^{0.400} 3x^2 \, dx = x^3\Big|_{0.300}^{0.400} = (0.400)^2 - (0.300)^2 = \boxed{0.037\ 0}$$

(c) The expectation value is

$$\langle x \rangle = \int_{\text{all space}} \psi^* x \psi \, dx = \int_0^{1.00} 3x^3 \, dx = \frac{3x^4}{4}\Big|_0^{1.00} = \boxed{0.750}$$

P41.4 The probability is given by

$$P = \int_{-a}^{a} |\psi(x)|^2 = \int_{-a}^{a} \frac{a}{\pi\left(x^2 + a^2\right)} dx = \left(\frac{a}{\pi}\right)\left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)\Big|_{-a}^{a}$$

$$P = \frac{1}{\pi}\left[\tan^{-1} 1 - \tan^{-1}(-1)\right] = \frac{1}{\pi}\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \boxed{\frac{1}{2}}$$

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Section 41.2 Analysis Model: Quantum Particle Under Boundary Conditions

P41.5 (a) The energy of a quantum particle confined to a line segment is

$$E_n = \frac{h^2 n^2}{8mL^2}$$

Here we have for the ground state

$$E_1 = \frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)^2 (1)^2}{8\left(1.67 \times 10^{-27}\text{kg}\right)\left(2.00 \times 10^{-14}\text{m}\right)^2}$$

$$= 8.22 \times 10^{-14}\text{J} = \boxed{0.513\ \text{MeV}}$$

and for the first and second excited states, which are states 2 and 3,

$$E_2 = 4E_1 = \boxed{2.05\ \text{MeV}} \quad \text{and} \quad E_3 = 9E_1 = \boxed{4.62\ \text{MeV}}$$

(b) They do; the MeV is the natural unit for energy radiated by an atomic nucleus.

Stated differently: Scattering experiments show that an atomic nucleus is a three-dimensional object always less than 15 fm in diameter. This one-dimensional box 20 fm long is a good model in energy terms.

P41.6 From Equation 41.14, the allowed energy levels of a particle in a box is

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2, \qquad n = 1,\ 2,\ 3,\ldots$$

(a) For $L = 1.00$ nm,

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2$$

$$= \left(\frac{1\ \text{eV}}{1.60 \times 10^{-19}\ \text{J}}\right)\left[\frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)^2}{8\left(9.11 \times 10^{-31}\ \text{kg}\right)\left(1.00 \times 10^{-9}\ \text{m}\right)^2}\right]n^2$$

$$= 0.377n^2 = 6\ \text{eV}$$

$$\boxed{n \approx 4}$$

(b) For $n = 4$, $E_n = 0.377(4)^2 = \boxed{6.03\ \text{eV}}$

P41.7 (a) From Equation 41.14, the allowed energy levels of an electron in a box is

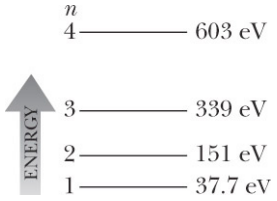
$$E_n = \left(\frac{h^2}{8m_e L^2}\right)n^2 \qquad n = 1,\ 2,\ 3,\ \ldots$$

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Substituting numerical values,

$$E_n = \left[\frac{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)^2}{8\left(9.11 \times 10^{-31}\ \text{kg}\right)\left(0.100 \times 10^{-9}\ \text{m}\right)^2} \right] n^2$$

$$= \left(6.02 \times 10^{-18}\ \text{J}\right) n^2 = (37.7\ \text{eV}) n^2$$



ANS. FIG. P41.7

(b) When the electron falls from higher level n_i to lower level n_f, it emits energy

$$\Delta E_n = \left(\frac{h^2}{8 m_e L^2} \right)\left(n_i^2 - n_f^2\right) = (37.7\ \text{eV})\left(n_i^2 - n_f^2\right)$$

by emitting a photon of wavelength

$$\lambda = \frac{hc}{\Delta E_n} = \frac{8 m_e c L^2}{h\left(n_i^2 - n_f^2\right)}$$

$$= \frac{8\left(9.109 \times 10^{-31}\ \text{kg}\right)\left(2.998 \times 10^{8}\ \text{m/s}\right)\left(0.100 \times 10^{-9}\ \text{m}\right)^2}{\left(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}\right)\left(n_i^2 - n_f^2\right)} \times \left(\frac{1\ \text{nm}}{10^{-9}\ \text{m}} \right)$$

$$= \frac{33.0\ \text{nm}}{\left(n_i^2 - n_f^2\right)}$$

For example, for the transition $4 \to 3$, the wavelength is

$$\lambda = \frac{33.0\ \text{nm}}{(4)^2 - (3)^2} = 4.71\ \text{nm}$$

The wavelengths produced by all possible transitions are:

Transition	$4 \to 3$	$4 \to 2$	$4 \to 1$	$3 \to 2$	$3 \to 1$	$2 \to 1$
λ (nm)	4.71	2.75	2.20	6.59	4.12	11.0

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P41.8 The energy of the photon is

$$E = \frac{hc}{\lambda} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{6.06\ \text{mm}}\left(\frac{1\ \text{mm}}{10^6\,\text{nm}}\right) = 2.05\times10^{-4}\,\text{eV}$$

The allowed energies of the proton in the box are

$$\begin{aligned} E_n &= \left(\frac{h^2}{8mL^2}\right)n^2 \\ &= \left[\frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{8\left(1.673\times10^{-27}\ \text{kg}\right)\left(1.00\times10^{-9}\ \text{m}\right)^2}\right]\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right)n^2 \\ &= \left(2.05\times10^{-4}\ \text{eV}\right)n^2 \end{aligned}$$

The smallest possible energy for a transition between states is from $n = 1$ to $n = 2$, which has energy

$$\Delta E_n = \left(2.05\times10^{-4}\,\text{eV}\right)\left(2^2-1^2\right) = 6.14\times10^{-4}\,\text{eV}$$

The photon does not have enough energy to cause this transition. The photon energy would be sufficient to cause a transition from $n = 0$ to $n = 1$, but the $n = 0$ state does not exist for the particle in a box.

P41.9 From Equation 41.14,

$$\Delta E = \frac{hc}{\lambda} = \left(\frac{h^2}{8m_eL^2}\right)\left[2^2-1^2\right] = \frac{3h^2}{8m_eL^2}$$

Solving for the length of the box then gives

$$\begin{aligned} L &= \sqrt{\frac{3h\lambda}{8m_ec}} \\ &= \sqrt{\frac{3\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)\left(694.3\times10^{-9}\ \text{m}\right)}{8\left(9.11\times10^{-31}\ \text{kg}\right)\left(3.00\times10^{8}\ \text{m/s}\right)}} \\ &= 7.95\times10^{-10}\ \text{m} = \boxed{0.795\ \text{nm}} \end{aligned}$$

P41.10 From Equation 41.14,

$$\Delta E = \frac{hc}{\lambda} = \left(\frac{h^2}{8m_eL^2}\right)\left[2^2-1^2\right] = \frac{3h^2}{8m_eL^2}$$

Solving for the length of the box then gives

$$L = \boxed{\sqrt{\frac{3h\lambda}{8m_ec}}}$$

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P41.11 From Equation 41.14, the allowed energy levels of a particle in a box is

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2 = n^2 E_1 \qquad n = 1, 2, 3, \ldots$$

For a proton ($m = 1.673 \times 10^{-27}$ kg) in a 10.0-fm wide box:

$$E_1 = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{8\left(1.673\times10^{-27}\ \text{kg}\right)\left(10.0\times10^{-15}\ \text{m}\right)^2}$$

$$= 3.28\times10^{-13}\ \text{J}\left(\frac{1\ \text{eV}}{1.602\times10^{-19}\ \text{J}}\right) = 2.05\times10^{6}\ \text{eV} = 2.05\ \text{MeV}$$

(a) The energy of the emitted photon is

$$E = \Delta E_n = E_2 - E_1 = (2)^2 E_1 - E_1 = 3E_1 = \boxed{6.14\ \text{MeV}}$$

(b) The wavelength of the photon is

$$\lambda = \frac{hc}{E} = \frac{1\ 240\ \text{eV}\cdot\text{nm}}{6.14\times10^{6}\ \text{eV}}$$

$$= 2.02\times10^{-4}\ \text{nm} = 2.02\times10^{-13}\ \text{m} = 202\times10^{-15}\ \text{m} = \boxed{202\ \text{fm}}$$

(c) This is a gamma ray, according to the electromagnetic spectrum chart in Chapter 34.

P41.12 The ground state energy of a particle (mass m) in a 1-dimensional box of width L is $E_1 = \dfrac{h^2}{8mL^2}$.

(a) For a proton ($m = 1.67 \times 10^{-27}$ kg) in a 0.200-nm wide box:

$$E_1 = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{8\left(1.67\times10^{-27}\ \text{kg}\right)\left(2.00\times10^{-10}\ \text{m}\right)^2}$$

$$= 8.22\times10^{-22}\ \text{J} = \boxed{5.13\times10^{-3}\ \text{eV}}$$

(b) For an electron ($m = 9.11 \times 10^{-31}$ kg) in the same size box:

$$E_1 = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{8\left(9.11\times10^{-31}\ \text{kg}\right)\left(2.00\times10^{-10}\ \text{m}\right)^2}$$

$$= 1.51\times10^{-18}\ \text{J} = \boxed{9.41\ \text{eV}}$$

(c) The electron has a much higher energy because it is much less massive.

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P41.13 $E_1 = 2.00 \text{ eV} = 3.20 \times 10^{-19} \text{ J}$. For the ground state,

$$E_1 = \frac{h^2}{8m_e L^2}$$

(a) The length of the region is

$$L = \frac{h}{\sqrt{8m_e E_1}} = \frac{6.626\times10^{-34}\ \text{J}\cdot\text{s}}{\sqrt{8\left(9.11\times10^{-31}\ \text{kg}\right)\left(3.20\times10^{-19}\ \text{J}\right)}}$$

$$= 4.34\times10^{-10}\ \text{m} = \boxed{0.434\ \text{nm}}$$

(b) For the excited states, $E_n = \left(\frac{h^2}{8m_e L^2}\right)n^2 = n^2 E_1$. For the first excited state, $\Delta E = E_2 - E_1 = 4E_1 - E_1 = 3E_1 = \boxed{6.00\ \text{eV}}$

P41.14 (a) The classical kinetic energy of the particle is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}\left(4.00\times10^{-3}\ \text{kg}\right)\left(1.00\times10^{-3}\ \text{m/s}\right)^2$$

$$= \boxed{2.00\times10^{-9}\ \text{J}}$$

(b) The length L can be found from

$$E = \left(\frac{h^2}{8mL^2}\right)n^2$$

Solving,

$$L = n\sqrt{\frac{h^2}{8mE}} = 2\sqrt{\frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{8\left(4.00\times10^{-3}\ \text{kg}\right)\left(2.00\times10^{-9}\ \text{J}\right)}}$$

$$= \boxed{1.66\times10^{-28}\ \text{m}}$$

(c) No. The length of the box would have to be much smaller than the size of a nucleus (~ 10^{-14} m) to confine the particle.

***P41.15** (a) The energies of the confined electron are $E_n = \frac{h^2}{8m_e L^2}n^2$. Its energy gain in the quantum jump from state 1 to state 4 is $\frac{h^2}{8m_e L^2}\left(4^2 - 1^2\right)$, and this is the photon energy $\frac{h^2 15}{8m_e L^2} = hf = \frac{hc}{\lambda}$.

Then $8m_e cL^2 = 15h\lambda$ and $\boxed{L = \left(\frac{15h\lambda}{8m_e c}\right)^{1/2}}$.

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(b) Let λ' represent the wavelength of the photon emitted:

$$\frac{hc}{\lambda'} = \frac{h^2}{8m_eL^2}4^2 - \frac{h^2}{8m_eL^2}2^2 = \frac{12h^2}{8m_eL^2}$$

Then $\frac{hc}{\lambda}\frac{\lambda'}{hc} = \frac{h^2 15(8m_eL^2)}{8m_eL^2 12h^2} = \frac{5}{4}$ and $\boxed{\lambda' = 1.25\lambda}$.

P41.16 (a) From $\Delta x \Delta p \geq \frac{\hbar}{2}$, with $\Delta x = L$, $\Delta p \geq \frac{\hbar}{2\Delta x} = \frac{\hbar}{2L}$, so the uncertainty in momentum must be at least $\Delta p \approx \boxed{\frac{\hbar}{2L}}$.

(b) Its energy is all kinetic, so

$$E = \frac{p^2}{2m} = \frac{(\Delta p)^2}{2m} \approx \boxed{\frac{\hbar^2}{8mL^2}} = \frac{h^2}{(4\pi)^2 8mL^2}$$

(c) Compare the result of part (b) to the result $h^2/8mL^2$ for the wave function as a standing wave. This estimate is too low by $4\pi^2 \approx 40$ times, but it correctly displays the pattern of dependence of the energy on the mass and on the length of the well.

P41.17 (a) $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$ becomes

$$A^2 \int_{-L/4}^{L/4} \cos^2\left(\frac{2\pi x}{L}\right) dx = A^2 \int_{-L/4}^{L/4} \frac{1 + \cos\left[2(2\pi x/L)\right]}{2} dx = 1$$

$$\frac{A^2}{2}\left[x + \frac{L}{4\pi}\cos\left(\frac{4\pi x/L}{L}\right)\right]\Bigg|_{-L/4}^{L/4} = 1$$

$$\frac{A^2}{2}\left(\frac{L}{2}\right) = 1 \rightarrow \boxed{A = \frac{2}{\sqrt{L}}}$$

(b) The probability of finding the particle between 0 and $\frac{L}{8}$ is

$$\int_0^{L/8} |\psi|^2 dx = A^2 \int_0^{L/8} \cos^2\left(\frac{2\pi x}{L}\right) dx = \frac{A^2}{2}\left[x + \frac{L}{4\pi}\cos\left(\frac{4\pi x}{L}\right)\right]\Bigg|_0^{L/8}$$

$$\frac{1}{2}\left(\frac{4}{L}\right)\left[\frac{L}{8} + \frac{L}{4\pi}\sin\left(\frac{\pi}{2}\right)\right] = \frac{1}{4} + \frac{1}{2\pi} = \boxed{0.409}$$

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P41.18 Normalization requires $\int_{\text{all space}} |\psi|^2 dx = 1$:

$$\int_0^L A^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = \int_0^L A^2 \frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2} dx = 1$$

$$= \frac{A^2}{2}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_0^L = 1$$

$$= \frac{A^2}{2}\left[L - \frac{L}{2\pi}\sin 2\pi\right]_0^L = \frac{A^2 L}{2} = 1$$

$$A = \sqrt{\frac{2}{L}}$$

P41.19 (a) The expectation value is $\langle x\rangle = \int_0^L \psi^* x\psi\, dx$:

$$\langle x\rangle = \int_0^L x\frac{2}{L}\sin^2\left(\frac{2\pi x}{L}\right) dx = \frac{2}{L}\int_0^L x\left\{\frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2}\right\} dx$$

$$= \frac{1}{L}\int_0^L x\left(1-\cos\frac{4\pi x}{L}\right) dx$$

From integral tables, we find that

$$\langle x\rangle = \frac{1}{L}\frac{x^2}{2}\bigg|_0^L - \frac{1}{L}\frac{L^2}{16\pi^2}\left[\frac{4\pi x}{L}\sin\frac{4\pi x}{L} + \cos\frac{4\pi x}{L}\right]_0^L = \boxed{\frac{L}{2}}$$

(b) The probability of finding the particle in the range $0.490L \le x \le 0.510L$ is

$$P = \frac{2}{L}\int_{0.490L}^{0.510L} \sin^2\left(\frac{2\pi x}{L}\right) dx = \frac{2}{L}\int_{0.490L}^{0.510L} \frac{1-\cos\left[2\left(2\pi x/L\right)\right]}{2} dx$$

$$= \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{4\pi x}{L}\right)\right]_{0.490L}^{0.510L}$$

$$= 0.020 - \frac{1}{4\pi}\left(\sin 2.04\pi - \sin 1.96\pi\right) = \boxed{5.26\times10^{-5}}$$

(c) The probability of finding the particle in the range $0.240L \le x \le 0.260L$ is

$$P = \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{4\pi x}{L}\right)\right]_{0.240L}^{0.260L} = \boxed{3.99\times10^{-2}}$$

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(d) In the $n = 2$ graph in the text's Figure 41.4(b), it is more probable to find the particle either near $x = L/4$ or $x = 3L/4$ than at the center, where the probability density is zero. Nevertheless, the symmetry of the distribution means that the average position is $x = L/2$.

P41.20 (a) The most probable positions of the particle are $x = L/4$, $L/2$, and $3L/4$.

(b) We look for sin $(3\pi x/L)$ taking on its extreme values 1 and –1 so that the squared wave function is as large as it can be. The result can also be found by studying Figure 41.4b. The most probable locations are at the antinodes of the standing wave pattern $n = 3$, which has three antinodes that are equally spaced, one at the center, and two a distance $L/4$ from either end.

P41.21 (a) The probability of finding the electron between $x = 0$ and $x = 0.100$ nm $= L/3$ is

$$\int_0^{L/3} |\psi_1|^2 \, dx = \frac{2}{L}\int_0^{L/3} \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{2}{L}\int_0^{L/3} \frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2} dx$$

$$= \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_0^{L/3}$$

$$= \frac{1}{3} - \frac{1}{2\pi}\sin\left(\frac{2\pi}{3}\right) = \frac{1}{3} - \frac{0.866}{2\pi} = \boxed{0.196}$$

(b) Classically, the particle moves back and forth steadily, spending equal time intervals in each third of the line. The classical probability is 0.333, which is significantly larger.

(c) The probability is

$$\int_0^{L/3} |\psi_{99}|^2 \, dx = \frac{2}{L}\int_0^{L/3} \sin^2\left(\frac{99\pi x}{L}\right) dx = \frac{1}{L}\int_0^{L/3} \left[1-\cos\left(\frac{198\pi x}{L}\right)\right] dx$$

$$= \frac{1}{L}\left[x - \frac{L}{198\pi}\sin\left(\frac{198\pi x}{L}\right)\right]_0^{L/3}$$

$$= \frac{1}{3} - \frac{1}{198\pi}\sin(66\pi) = \frac{1}{3} - 0 = 0.333$$

The probability is 0.333 for both classical and quantum models.

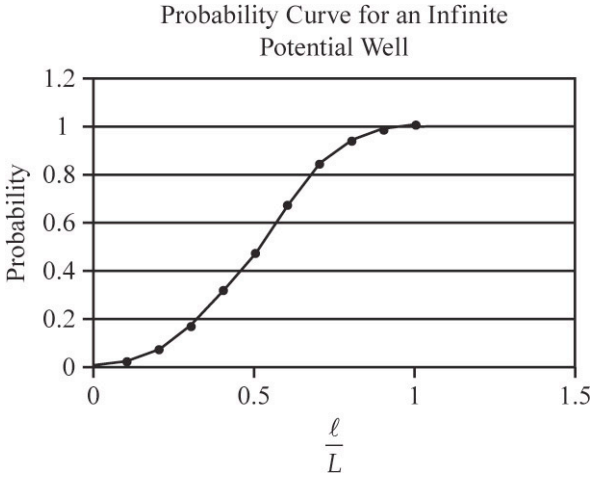
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P41.22 (a) From Equation 41.13, $\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$. The probability of finding the particle between $x = 0$ and $x = \ell$ is

$$\int_0^{\ell} |\psi_1|^2 \, dx = \frac{2}{L}\int_0^{\ell} \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{2}{L}\int_0^{\ell} \frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2} dx$$

$$= \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]\Bigg|_0^{\ell} = \boxed{\frac{\ell}{L} - \frac{1}{2\pi}\sin\left(\frac{2\pi \ell}{L}\right)}$$

(b) The probability function is sketched in ANS. FIG. P41.22(b).



ANS. FIG. P41.22(b)

(c) The wave function is zero for $x < 0$ and for $x > L$. The probability at $\ell = 0$ must be zero because the particle is never found at $x < 0$ or exactly at $x = 0$. The probability at $\ell = L$ must be 1 for normalization: the particle is always found somewhere in the range $0 < x < L$.

(d) The probability of finding the particle between $x = 0$ and $x = \ell$ is $\frac{2}{3}$, and between $x = \ell$ and $x = L$ is $\frac{1}{3}$.

Thus, $$\int_0^{\ell} |\psi_1|^2 \, dx = \frac{2}{3}$$

$$\therefore \frac{\ell}{L} - \frac{1}{2\pi}\sin\left(\frac{2\pi \ell}{L}\right) = \frac{2}{3}$$

or, defining $u = \frac{\ell}{L}$, $$u - \frac{1}{2\pi}\sin 2\pi u = \frac{2}{3}$$

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This equation for u can be solved by homing in on the solution with a calculator, the result being $u = \frac{\ell}{L} = 0.585$, or $\ell = \boxed{0.585L}$ to three digits.

P41.23 (a) The probability is

$$P = \int_0^{L/3} |\psi_1|^2 \, dx = \frac{2}{L}\int_0^{L/3} \sin^2\left(\frac{\pi x}{L}\right) dx$$

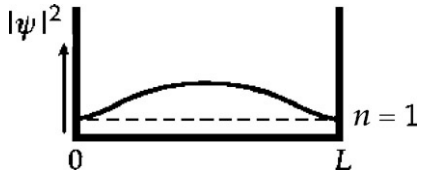
$$= \frac{2}{L}\int_0^{L/3} \frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2} dx$$

$$= \frac{1}{L}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]\Bigg|_0^{L/3} = \frac{1}{L}\left[\frac{L}{3} - \frac{L}{2\pi}\sin\left(\frac{2\pi}{3}\right)\right]$$

$$= \frac{1}{3} - \frac{1}{2\pi}\sin\left(\frac{2\pi}{3}\right)$$

$$= \left(\frac{1}{3} - \frac{\sqrt{3}}{4\pi}\right) = \boxed{0.196}$$

(b) The probability density is symmetric about $x = \frac{L}{2}$. Thus, the probability of finding the particle between $x = \frac{2L}{3}$ and $x = L$ is the same, 0.196. Therefore, the probability of finding it in the range $\frac{L}{3} \le x \le \frac{2L}{3}$ is $P = 1.00 - 2(0.196) = \boxed{0.609}$.



ANS. FIG. P41.23(b)

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Section 41.3 The Schrödinger Equation

P41.24 From $\psi = Ae^{i(kx - \omega t)}$ **[1]**

we evaluate

$$\frac{d\psi}{dx} = ikAe^{i(kx - \omega t)}$$

and $$\frac{d^2\psi}{dx^2} = -k^2 Ae^{i(kx - \omega t)}$$ **[2]**

We substitute equations [1] and [2] into the Schrödinger equation, so that Equation 41.15,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi$$

becomes the test equation

$$\left(-\frac{\hbar^2}{2m}\right)\left(-k^2 Ae^{i(kx - \omega t)}\right) + 0 = EAe^{i(kx - \omega t)}$$ **[3]**

The wave function $\psi = Ae^{i(kx - \omega t)}$ is a solution to the Schrödinger equation if equation [3] is true. Both sides depend on A, x, and t in the same way, so we can cancel several factors, and determine that we have a solution if

$$\frac{\hbar^2 k^2}{2m} = E$$

But this is true for a nonrelativistic particle with mass in a region where the potential energy is zero, since

$$\frac{\hbar^2 k^2}{2m} = \frac{1}{2m}\left(\frac{h}{2\pi}\right)^2\left(\frac{2\pi}{\lambda}\right)^2 = \underbrace{\frac{(h/\lambda)^2}{2m} = \frac{p^2}{2m}}_{\text{using de Broglie's equation}}$$

$$= \frac{m^2u^2}{2m} = \frac{1}{2}mu^2 = \underbrace{K = K + U}_{\text{recall } U=0} = E$$

where K is the kinetic energy. Therefore, the given wave function does satisfy Equation 41.15.

P41.25 (a) Given the function

$$\psi(x) = A\cos kx + B\sin kx$$

Its derivative with respect to x is

$$\frac{\partial\psi}{\partial x} = -kA\sin kx + kB\cos kx$$

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And its second derivative is

$$\frac{\partial^2\psi}{\partial x^2} = -k^2 A\cos kx - k^2 B\sin kx$$
$$= -k^2\left(A\cos kx + B\sin kx\right) = -k^2\psi$$

The Schrödinger equation is satisfied if

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi, \qquad \text{where} \quad U = 0:$$

$$-\frac{\hbar^2}{2m}\left(-k^2\psi\right) = E\psi \qquad \rightarrow \qquad \frac{\hbar^2 k^2}{2m}\psi = E\psi$$

This is true as an identity (functional equality) for all x if $E = \dfrac{\hbar^2 k^2}{2m}$, which is true because $E = K + U = K + 0 = K$, and

$$\frac{\hbar^2 k^2}{2m} = \frac{1}{2m}\left(\frac{h}{2\pi}\right)^2\left(\frac{2\pi}{\lambda}\right)^2 = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2 = \frac{p^2}{2m} = K$$

(b) From part (a), $\boxed{E = \dfrac{\hbar^2 k^2}{2m}}$.

P41.26 (a) These are standing wave patterns with nodes at the ends and n antinodes.

For $n = 1$, the wave function is

$$\psi_1(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{\pi x}{L}\right)$$

and the probability density is

$$P_1(x) = \left|\psi_1(x)\right|^2 = \frac{2}{L}\cos^2\left(\frac{\pi x}{L}\right)$$

For $n = 2$, the wave function is

$$\psi_2(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right)$$

and the probability density is

$$P_2(x) = \left|\psi_2(x)\right|^2 = \frac{2}{L}\sin^2\left(\frac{2\pi x}{L}\right)$$

For $n = 3$, the wave function is

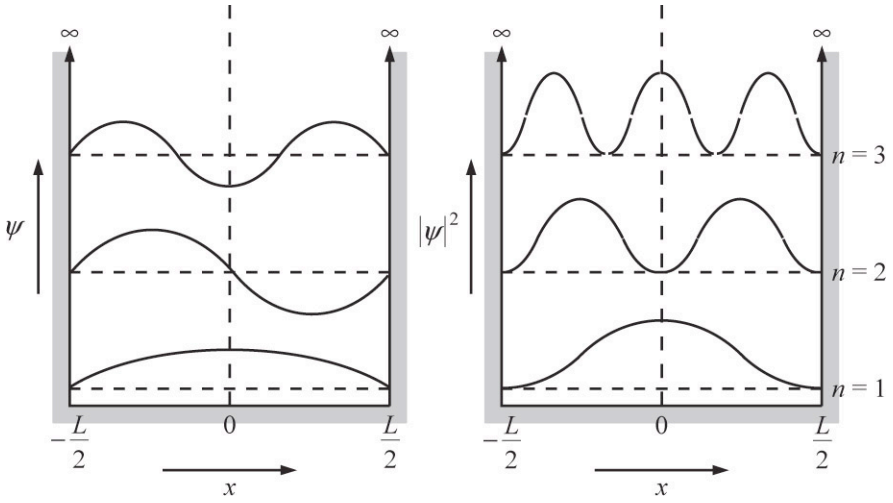
$$\psi_3(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{3\pi x}{L}\right)$$

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and the probability density is

$$P_3(x) = |\psi_3(x)|^2 = \frac{2}{L}\cos^2\left(\frac{3\pi x}{L}\right)$$

(b) The wave functions and probability densities are shown in ANS. FIG. P41.26(b).



ANS. FIG. P41.26(b)

P41.27 (a) Setting the total energy E equal to zero and rearranging the Schrödinger equation to isolate the potential energy function gives

$$\left(\frac{\hbar^2}{2m}\right)\frac{d^2\psi}{dx^2} + U(x)\psi = 0$$

$$U(x) = \left(\frac{\hbar^2}{2m}\right)\frac{1}{\psi}\frac{d^2\psi}{dx^2}$$

If $\psi(x) = Axe^{-x^2/L^2}$

Then,

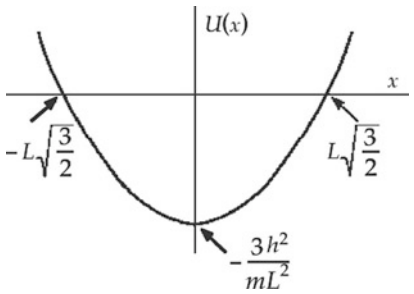
$$\frac{d^2\psi}{dx^2} = \left(4Ax^3 - 6AxL^2\right)\frac{e^{-x^2/L^2}}{L^4}$$

or $$\frac{d^2\psi}{dx^2} = \frac{\left(4x^2 - 6L^2\right)}{L^4}\psi(x)$$

and $$\boxed{U(x) = \frac{\hbar^2}{2mL^2}\left(\frac{4x^2}{L^2} - 6\right)}$$

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(b) U(x) is sketched in ANS. FIG. P41.27(b).



ANS. FIG. P41.27(b)

P41.28 (a) $\psi(x) = A\left(1 - \frac{x^2}{L^2}\right) \rightarrow \frac{d\psi}{dx} = -\frac{2Ax}{L^2} \rightarrow \frac{d^2\psi}{dx^2} = -\frac{2A}{L^2}$

Schrödinger's equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi$$

becomes

$$-\frac{\hbar^2}{2m}\left(-\frac{2A}{L^2}\right) + \frac{\left(-\hbar^2 x^2\right)}{mL^2\left(L^2 - x^2\right)}A\left(1 - \frac{x^2}{L^2}\right) = EA\left(1 - \frac{x^2}{L^2}\right)$$

$$-\frac{\hbar^2}{\cancel{2}m}\left(-\frac{\cancel{2}}{L^2}\right) + \frac{\left(-\hbar^2 x^2\right)\cancel{\left(L^2 - x^2\right)}}{mL^4\cancel{\left(L^2 - x^2\right)}} = E\left(1 - \frac{x^2}{L^2}\right)$$

$$\frac{\hbar^2}{mL^2} + \frac{\left(-\hbar^2 x^2\right)}{mL^4} = E\left(1 - \frac{x^2}{L^2}\right)$$

$$\frac{\hbar^2}{mL^2}\left(1 - \frac{x^2}{L^4}\right) = E\left(1 - \frac{x^2}{L^2}\right)$$

This will be true for all x if $\boxed{E = \frac{\hbar^2}{L^2 m}}$.

(b) Note that the wave function $\psi(x)$ is an even function; therefore, we may write the normalization condition as

$$\int_{-L}^{L} |\psi|^2 dx = 1 = \int_{-L}^{L} A^2\left(1 - \frac{x^2}{L^2}\right)^2 dx = 2\int_{0}^{L} A^2\left(1 - \frac{x^2}{L^2}\right)^2 dx$$

$$= 2A^2\int_{0}^{L}\left(1 - \frac{2x^2}{L^2} + \frac{x^4}{L^4}\right)dx$$

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Solving,

$$1 = 2A^2\left[x - \frac{2x^3}{3L^2} + \frac{x^5}{5L^4}\right]_0^L = 2A^2\left[L - \frac{2}{3}L + \frac{L}{5}\right]$$

$$= A^2\left(\frac{16L}{15}\right) \quad \rightarrow \quad \boxed{A = \sqrt{\frac{15}{16L}}}$$

(c) As in part (b), because the wave function is an even function, the probability is

$$P = \int_{-L/3}^{L/3} \psi^2\,dx = \int_0^{L/3} \psi^2\,dx = 2\frac{15}{16L}\int_0^{L/3}\left(1 - \frac{2x^2}{L^2} + \frac{x^4}{L^4}\right)dx$$

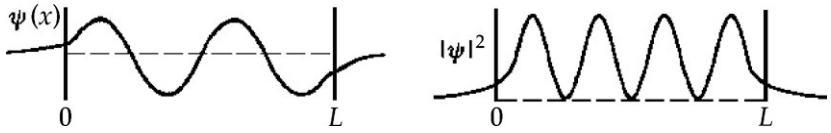
$$= \frac{15}{8L}\left[x - \frac{2x^3}{3L^2} + \frac{x^5}{5L^5}\right]_0^{L/3}$$

$$= \frac{15}{8L}\left[\frac{L}{3} - \frac{2L}{81} + \frac{L}{1215}\right] = \frac{47}{81} = \boxed{0.580}$$

Section 41.4 A Particle in a Well of Finite Height

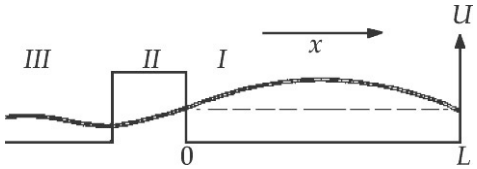
P41.29 (a) For $n = 4$, the wave function has two maxima and two minima (four extrema), as shown in the left-hand panel of ANS. FIG. P41.29.

(b) For $n = 4$, the probability function has four maxima. as shown in the right-hand panel of ANS. FIG. P41.29.



ANS. FIG. P41.29

P41.30 (a) See ANS. FIG. P41.30(a).



ANS. FIG. P41.30(a)

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(b) The wavelength inside the box is $2L$. The wave function penetrates the wall, but the wavelength of the transmitted wave traveling to the left is the same, $\boxed{2L}$, because $U = 0$ on both sides of the wall, so the energy and momentum and, therefore, the wavelength, are the same.

Section 41.5 Tunneling Through a Potential Energy Barrier

P41.31 The decay constant for the wave function inside the barrier is:

$$C = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$= \frac{\sqrt{2\left(9.11\times10^{-31}\ \text{kg}\right)(10.0\ \text{eV} - 5.00\ \text{eV})\left(1.60\times10^{-19}\ \text{J/eV}\right)}}{6.626\times10^{-34}\ \text{J}\cdot\text{s}/2\pi}$$

$$= 1.14\times10^{10}\ \text{m}^{-1}$$

(a) The approximate probability of transmission is

$$T \approx e^{-2CL} = e^{-2\left(1.14\times10^{10}\ \text{m}^{-1}\right)\left(2.00\times10^{-10}\ \text{m}\right)} = \boxed{0.010\ 3}$$

or a 1% chance of transmission.

(b) $R = 1 - T = \boxed{0.990}$, a 99% chance of reflection.

P41.32 (a) $T = e^{-2CL}$, where

$$C = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$= \frac{\sqrt{2\left(9.11\times10^{-31}\ \text{kg}\right)(5.00-4.50)\left(1.60\times10^{-19}\ \text{J}\right)}}{1.055\times10^{-34}\ \text{J}\cdot\text{s}}$$

$$= 3.62\times10^{9}\ \text{m}^{-1}$$

and $T = e^{-2CL} = \exp\left[-2\left(3.62\times10^{9}\ \text{m}^{-1}\right)\left(950\times10^{-12}\ \text{m}\right)\right]$

$$= \exp(-6.88) = \boxed{1.03\times10^{-3}}$$

(b) We require $e^{-2CL} = 10^{-6}$. Taking logarithms,

$$-2CL = \ln 10^{-6} = -6\ln 10$$

$$L = \frac{3\ln 10}{C} = \frac{3\ln 10}{3.62\times10^{9}\ \text{m}^{-1}} = 1.91\times10^{-9}\ \text{m} = \boxed{1.91\ \text{nm}}$$

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P41.33 The original tunneling probability is $T = e^{-2CL}$, where

$$C = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$= \frac{\sqrt{2\left(9.11\times10^{-31}\text{ kg}\right)(20.0-12.0)\left(1.60\times10^{-19}\text{ J}\right)}}{6.626\times10^{-34}\text{ J}\cdot\text{s}/2\pi}$$

$$= 1.448\ 1\times10^{10}\text{ m}^{-1}$$

The photon energy is $hf = \frac{hc}{\lambda} = \frac{1\,240\text{ eV}\cdot\text{nm}}{546\text{ nm}} = 2.27\text{ eV}$, to make the electron's new kinetic energy 12.0 + 2.27 = 14.27 eV and its decay coefficient inside the barrier

$$C' = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$= \frac{\sqrt{2\left(9.11\times10^{-31}\text{ kg}\right)(20.0-14.27)\left(1.60\times10^{-19}\text{ J}\right)}}{6.626\times10^{-34}\text{ J}\cdot\text{s}/2\pi}$$

$$= 1.225\ 5\times10^{10}\text{ m}^{-1}$$

Now the factor of increase in transmission probability is

$$\frac{e^{-2C'L}}{e^{-2CL}} = e^{2L(C-C')} = e^{2\left(1.00\times10^{-9}\text{ m}\right)\left(0.223\times10^{10}\text{ m}^{-1}\right)} = e^{4.45} = \boxed{85.9}$$

Section 41.6 Applications of Tunneling

P41.34 With the wave function proportional to e^{-CL}, the transmission coefficient and the tunneling current are proportional to $|\psi|^2$, to e^{-2CL}. Then,

$$\frac{I(0.500\text{ nm})}{I(0.515\text{ nm})} = \frac{e^{-2(10.0/\text{nm})(0.500\text{ nm})}}{e^{-2(10.0/\text{nm})(0.515\text{ nm})}} = e^{20.0(0.015)} = \boxed{1.35}$$

P41.35 With transmission coefficient e^{-2CL}, the fractional change in transmission is

$$\frac{e^{-2(10.0/\text{nm})L} - e^{-2(10.0/\text{nm})(L+0.002\,00\text{ nm})}}{e^{-2(10.0/\text{nm})L}} = 1 - e^{-20.0(0.002\,00)}$$

$$= 0.039\ 2 = \boxed{3.92\%}$$

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Section 41.7 The Simple Harmonic Oscillator

P41.36 (a) The wave function is given by $\psi = Axe^{-bx^2}$, so

$$\frac{d\psi}{dx} = Ae^{-bx^2} - 2bx^2Ae^{-bx^2}$$

and

$$\frac{d^2\psi}{dx^2} = \left[-2bxAe^{-bx^2} - 4bxAe^{-bx^2}\right] + 4b^2x^3e^{-bx^2} = -6b\psi + 4b^2x^2\psi$$

Substitute into Equation 41.24:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi$$

$$-\frac{\hbar^2}{2m}\left[-6b\psi + 4b^2x^2\psi\right] + \frac{1}{2}m\omega^2x^2\psi = E\psi$$

$$\frac{3b\hbar^2}{m}\psi - \frac{2b^2\hbar^2}{m}x^2\psi = -\frac{1}{2}m\omega^2x^2\psi + E\psi$$

For this to be true as an identity, the coefficients of like terms must be the same for all values of x. So we must have both

$$\frac{2b^2\hbar^2}{m} = \frac{1}{2}m\omega^2 \quad \rightarrow \quad b^2 = \frac{m^2\omega^2}{4\hbar^2} \quad \text{and} \quad \frac{3b\hbar^2}{m} = E$$

(b) Therefore, $\boxed{b = \frac{m\omega}{2\hbar}}$ and $E = \frac{3b\hbar^2}{m} = \boxed{\frac{3}{2}\hbar\omega}$

(c) The energy levels are $E_n = \left(n + \frac{1}{2}\right)\hbar\omega = \frac{3}{2}\hbar\omega$, so $n = 1$, which corresponds to the $\boxed{\text{first excited state}}$.

P41.37 The longest wavelength corresponds to minimum photon energy, which must be equal to the spacing between energy levels of the oscillator. From $E = \hbar\omega$, we have

$$\frac{hc}{\lambda} = \hbar\sqrt{\frac{k}{m}} = \frac{h}{2\pi}\sqrt{\frac{k}{m}}$$

or

$$\lambda = 2\pi c\sqrt{\frac{m}{k}} = 2\pi\left(3.00\times10^8\ \text{m/s}\right)\left(\frac{9.11\times10^{-31}\ \text{kg}}{8.99\ \text{N/m}}\right)^{1/2}$$

$$= \boxed{600\ \text{nm}}$$

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P41.38 The longest wavelength corresponds to minimum photon energy, which must be equal to the spacing between energy levels of the oscillator, which is (from Equation 41.28)

$$E = \hbar\omega$$

$$\frac{hc}{\lambda} = \hbar\sqrt{\frac{k}{m}} = \frac{h}{2\pi}\sqrt{\frac{k}{m}}$$

$$\lambda = \boxed{2\pi c\sqrt{\frac{m}{k}}}$$

P41.39 (a) With $\psi = Be^{-(m\omega/2\hbar)x^2}$, the normalization condition $\int_{\text{all } x} |\psi|^2\, dx = 1$ becomes

$$1 = \int_{-\infty}^{\infty} B^2 e^{-2(m\omega/2\hbar)x^2}\, dx = 2B^2 \int_0^{\infty} e^{-(m\omega/\hbar)x^2}\, dx$$

$$= 2B^2 \frac{1}{2}\sqrt{\frac{\pi}{m\omega/\hbar}} = B^2\sqrt{\frac{\pi\hbar}{m\omega}}$$

where Table B.6 in Appendix B was used to evaluate the integral.

Thus, $\boxed{B = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}}$.

(b) For small δ, the probability of finding the particle in the range $-\frac{\delta}{2} < x < \frac{\delta}{2}$ is

$$\int_{-\delta/2}^{\delta/2} |\psi|^2\, dx \approx \delta|\psi(0)|^2 = \delta B^2 e^{-0} = \boxed{\delta\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}}$$

P41.40 (a) For the center of mass to be fixed, $m_1u_1 + m_2u_2 = 0$. Then

$$u = |u_1| + |u_2| = |u_1| + \frac{m_1}{m_2}|u_1| = \frac{m_2 + m_1}{m_2}|u_1|$$

and

$$|u_1| = \frac{m_2 u}{m_1 + m_2}$$

Also,

$$u = \frac{m_2}{m_1}|u_2| + |u_2| = \left(\frac{m_2 + m_1}{m_1}\right)|u_2| \quad \rightarrow \quad |u_2| = \frac{m_1 u}{m_1 + m_2}$$

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Substitute for $|u_1|$ and $|u_2|$:

$$\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2+\frac{1}{2}kx^2=\frac{1}{2}\frac{m_1m_2^2u^2}{(m_1+m_2)^2}+\frac{1}{2}\frac{m_2m_1^2u^2}{(m_1+m_2)^2}+\frac{1}{2}kx^2$$

$$=\frac{1}{2}\frac{m_1m_2(m_1+m_2)}{(m_1+m_2)^2}u^2+\frac{1}{2}kx^2$$

$$=\frac{1}{2}\mu u^2+\frac{1}{2}kx^2$$

(b) Because the total energy is constant

$$\frac{d}{dx}\left(\frac{1}{2}\mu u^2+\frac{1}{2}kx^2\right)=0$$

$$0=\frac{1}{2}\mu 2u\frac{du}{dx}+\frac{1}{2}k2x=\mu\frac{dx}{dt}\frac{du}{dx}+kx=\mu\frac{du}{dt}+kx=\mu a+kx$$

$$\mu a=-kx$$

$$a=-\frac{kx}{\mu}$$

This is the condition for simple harmonic motion; the acceleration of the equivalent particle is a negative constant times the displacement from equilibrium.

(c) By identification with $a=-\omega^2x$,

$$\omega=\sqrt{\frac{k}{\mu}}=2\pi f \text{ and } \boxed{f=\frac{1}{2\pi}\sqrt{\frac{k}{\mu}}}$$

P41.41 (a) With $\langle x\rangle=0$ and $\langle p_x\rangle=0$, the average value of x^2 is $(\Delta x)^2$ and the average value of p_x^2 is $(\Delta p_x)^2$. We know $\Delta x\geq\frac{\hbar}{2\Delta p_x}$.

The average of the energy is constant:

$$\langle E\rangle=\left\langle\frac{p_x^2}{2m}\right\rangle+\left\langle\frac{k}{2}x^2\right\rangle=\frac{\langle p_x^2\rangle}{2m}+\frac{k}{2}\langle x^2\rangle$$

$$E=\frac{(\Delta p_x)^2}{2m}+\frac{k}{2}(\Delta x)^2\geq\frac{(\Delta p_x)^2}{2m}+\frac{k}{2}\left(\frac{\hbar}{2\Delta p_x}\right)^2$$

$$E\geq\frac{(\Delta p_x)^2}{2m}+\frac{k\hbar^2}{8(\Delta p_x)^2}$$

We rewrite the last equation as $$E\geq\frac{p_x^2}{2m}+\frac{k\hbar^2}{8p_x^2}$$

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(b) To minimize E as a function of $(\Delta p_x)^2$, we require

$$\frac{d}{d\left[(\Delta p_x)^2\right]}\left[\frac{(\Delta p_x)^2}{2m}+\frac{k\hbar^2}{8(\Delta p_x)^2}\right]=0$$

$$\frac{1}{2m}+\frac{k\hbar^2}{8}(-1)\frac{1}{(\Delta p_x)^4}=0$$

Then

$$\frac{k\hbar^2}{8(\Delta p_x)^4}=\frac{1}{2m} \rightarrow (\Delta p_x)^2=\left(\frac{2mk\hbar^2}{8}\right)^{1/2}=\frac{\hbar\sqrt{mk}}{2}$$

and

$$E \geq \frac{(\Delta p_x)^2}{2m}+\frac{k\hbar^2}{8(\Delta p_x)^2}=\frac{\hbar\sqrt{mk}}{2(2m)}+\frac{k\hbar^2 2}{8\hbar\sqrt{mk}}$$

$$=\frac{\hbar}{4}\sqrt{\frac{k}{m}}+\frac{\hbar}{4}\sqrt{\frac{k}{m}}=\frac{\hbar}{2}\sqrt{\frac{k}{m}}$$

Therefore, $E_{\min}=\frac{\hbar}{2}\sqrt{\frac{k}{m}}=\boxed{\frac{\hbar\omega}{2}}$

P41.42 Equation 41.26 is $\psi = Be^{-(m\omega/2\hbar)x^2}$, so

$$\frac{d\psi}{dx}=-\left(\frac{m\omega}{\hbar}\right)x\psi \quad \text{and} \quad \frac{d^2\psi}{dx^2}=\left(\frac{m\omega}{\hbar}\right)^2 x^2\psi+\left(-\frac{m\omega}{\hbar}\right)\psi$$

Substitute into Equation 41.24:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+\frac{1}{2}m\omega^2x^2\psi=E\psi$$

$$-\frac{\hbar^2}{2m}\left[\left(\frac{m\omega}{\hbar}\right)^2 x^2\psi+\left(-\frac{m\omega}{\hbar}\right)\psi\right]+\frac{1}{2}m\omega^2x^2\psi=E\psi$$

$$\cancel{-\frac{1}{2}m\omega^2x^2\psi}+\left(\frac{\hbar\omega}{2}\right)\psi+\cancel{\frac{1}{2}m\omega^2x^2\psi}=E\psi$$

$$\left(\frac{\hbar\omega}{2}\right)\psi=E\psi$$

which is satisfied provided that $E=\frac{\hbar\omega}{2}$.

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Additional Problems

P41.43 (a) The particle's wavelength is

$$\lambda = \frac{2L}{n} = \frac{2L}{1} = \boxed{2.00 \times 10^{-10}\ \text{m}}$$

(b) Its momentum is

$$p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{2.00 \times 10^{-10}\ \text{m}} = \boxed{3.31 \times 10^{-24}\ \text{kg} \cdot \text{m/s}}$$

(c) And its energy is

$$E = \frac{p^2}{2m} = \frac{\left(3.31 \times 10^{-24}\ \text{kg} \cdot \text{m/s}\right)^2}{2\left(2.00 \times 10^{-28}\ \text{kg}\right)} = \boxed{0.171\ \text{eV}}$$

P41.44 (a) From Equation 41.4 for $\psi(x) = Ae^{ikx}$, the first and second derivatives are

$$\frac{d}{dx}\left(Ae^{ikx}\right) = ikAe^{ikx} \quad \text{and} \quad \frac{d^2\psi}{dx^2} = -k^2 Ae^{ikx}$$

Then

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = -\frac{\hbar^2}{2m}\left(-k^2 Ae^{ikx}\right) = \frac{\hbar^2 k^2}{2m}\left(Ae^{ikx}\right)$$

$$= \frac{1}{2m}\left(\frac{h}{\cancel{2\pi}}\right)^2\left(\frac{\cancel{2\pi}}{\lambda}\right)^2\left(Ae^{ikx}\right)$$

$$= \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2\left(Ae^{ikx}\right) = \frac{p^2}{2m}\psi = K\psi$$

(b) For $\psi(x) = A\sin\left(\frac{2\pi x}{\lambda}\right) = A\sin kx$,

$$\frac{d}{dx}(A\sin kx) = Ak\cos kx \quad \text{and} \quad \frac{d^2\psi}{dx^2} = -Ak^2\sin kx.$$

Then, similarly to the proof in part (a),

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = -\frac{\hbar^2}{2m}\left(-Ak^2\sin kx\right) = \frac{\hbar^2 k^2}{2m}\left(Ak^2\sin kx\right) = \frac{p^2}{2m}\psi$$

$$= K\psi$$

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P41.45 From Equation 41.13, $\psi_2(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right)$.

The probability of finding the particle between $x = 0$ and $x = L/4$ is

$$\int_0^{L/4} |\psi|^2\, dx = \int_0^{L/4} \frac{2}{L}\sin^2\left(\frac{2\pi x}{L}\right) dx = \frac{\cancel{2}}{L}\int_0^{L/4} \frac{1-\cos\left[2\left(2\pi x/L\right)\right]}{\cancel{2}}\, dx$$

$$\frac{1}{L}\left[x - \frac{L}{4\pi}\sin\left(\frac{4\pi x}{L}\right)\right]\Bigg|_0^{L/4} = \frac{1}{L}\left[\frac{L}{4} - \frac{L}{4\pi}\sin(\pi)\right] = \frac{1}{4} = \boxed{0.250}$$

P41.46 If we had $n = 0$ for a quantum particle in a box, its momentum would be zero. The uncertainty in its momentum would be zero. The uncertainty in its position would not be infinite, but just equal to the width of the box. Then the uncertainty product would be zero, to violate the uncertainty principle. The contradiction shows that the quantum number cannot be zero. In its ground state the particle has some nonzero zero-point energy.

P41.47 $T = e^{-2CL}$, where $C = \frac{\sqrt{2m(U-E)}}{\hbar}$ and where m is in kilograms, and U and E are in joules.

(a) We compute

$$C = \frac{\sqrt{2\left(9.11\times10^{-31}\text{ kg}\right)\left[\left(0.010\ 0\text{ eV}\right)\left(1.60\times10^{-19}\text{ J/eV}\right)\right]}}{1.055\times10^{-34}\text{ J}\cdot\text{s}}$$

$$= 5.12\times10^{8}\text{ m}^{-1}$$

Then,

$$2CL = 2\left(5.12\times10^{8}\text{ m}^{-1}\right)\left(0.100\times10^{-9}\text{ m}\right) = 0.102$$

and $T = e^{-0.102} = \boxed{0.903}$

(b) We compute

$$C = \frac{\sqrt{2\left(9.11\times10^{-31}\text{ kg}\right)\left[\left(1.00\text{ eV}\right)\left(1.60\times10^{-19}\text{ J/eV}\right)\right]}}{1.055\times10^{-34}\text{ J}\cdot\text{s}}$$

$$= 5.12\times10^{9}\text{ m}^{-1}$$

Then,

$$2CL = 2\left(5.12\times10^{9}\text{ m}^{-1}\right)\left(0.100\times10^{-9}\text{ m}\right) = 1.02$$

and $T = e^{-1.02} = \boxed{0.359}$

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(c) We compute

$$C = \frac{\sqrt{2\left(6.65\times10^{-27}\ \text{kg}\right)\left[\left(1.00\times10^{6}\ \text{eV}\right)\left(1.60\times10^{-19}\ \text{J/eV}\right)\right]}}{1.055\times10^{-34}\ \text{J}\cdot\text{s}}$$

$$= 4.37\times10^{14}\ \text{m}^{-1}$$

Then,

$$2CL = 2\left(4.37\times10^{14}\ \text{m}^{-1}\right)\left(1.00\times10^{-15}\ \text{m}\right) = 0.875$$

and $T = e^{-0.875} = \boxed{0.417}$

(d) We compute

$$2CL = 2\frac{\sqrt{2\left(8.00\ \text{kg}\right)\left(1.00\ \text{J}\right)}}{1.055\times10^{-34}\ \text{J}\cdot\text{s}}\left(0.020\ 0\ \text{m}\right) = 1.52\times10^{33}$$

Then,

$$T = e^{-1.52\times10^{33}} = e^{(\ln 10)(-1.52\times10^{33}/\ln 10)} = \boxed{10^{-6.59\times10^{32}}}$$

P41.48 From Equation 41.14, the energy levels of an electron in an infinitely deep potential well are proportional to n^2. If the energy of the ground state, $n = 1$, is $E_1 = 0.300$ eV, the energy levels of the states $n = 2$, 3, and 4 are

$$E_2 = 2^2\left(0.300\ \text{eV}\right) = 1.20\ \text{eV}$$

$$E_3 = 3^2\left(0.300\ \text{eV}\right) = 2.70\ \text{eV}$$

$$E_4 = 4^2\left(0.300\ \text{eV}\right) = 4.80\ \text{eV}$$

(a) For the transition from the $n = 3$ level to the $n = 1$ level, the electron loses energy

$$E = \frac{hc}{\lambda} = E_3 - E_1 = 2.70\ \text{eV} - 0.300\ \text{eV} = 2.40\ \text{eV}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\ \text{eV}\cdot\text{nm}}{2.40\ \text{eV}} = 517\ \text{nm}$$

(b) For the transition from level 2 to level 1,

$$E = 1.20\ \text{eV} - 0.300\ \text{eV} = 0.900\ \text{eV}$$

and

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\ \text{eV}\cdot\text{nm}}{0.900\ \text{eV}} = 1\ 380\ \text{nm} = \boxed{1.38\ \mu\text{m}}$$

This photon, with wavelength greater than 700 nm, is in the infrared region.

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In like manner, we find

for 3 to 2: $\Delta E = 1.50$ eV, and $\lambda =$ 827 nm, infrared

for 4 to 1: $\Delta E = 4.50$ eV, and $\lambda =$ 275 nm, ultraviolet

for 4 to 2: $\Delta E = 3.60$ eV, and $\lambda =$ 344 nm, near ultraviolet

for 4 to 3: $\Delta E = 2.10$ eV, and $\lambda =$ 590 nm, yellow-orange visible

P41.49 (a) From $E = hf$, the frequency is

$$f = \frac{E}{h} = \frac{(1.80\text{ eV})}{\left(6.626\times10^{-34}\text{ J}\cdot\text{s}\right)}\left(\frac{1.602\times10^{-19}\text{ J}}{1.00\text{ eV}}\right)$$

$$= 4.35\times10^{14}\text{ Hz} = 435\times10^{12}\text{ Hz} = \boxed{435\text{ THz}}$$

(b) The wavelength of the emitted photon is

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\text{ m/s}}{4.35\times10^{14}\text{ Hz}} = 6.89\times10^{-7}\text{ m} = \boxed{689\text{ nm}}$$

(c) We use $\Delta E\Delta t \geq \frac{\hbar}{2}$, so

$$\Delta E \geq \frac{\hbar}{2\Delta t} = \frac{h}{4\pi(\Delta t)} = \frac{6.626\times10^{-34}\text{ J}\cdot\text{s}}{4\pi\left(2.00\times10^{-6}\text{ s}\right)}$$

$$\Delta E \geq 2.64\times10^{-29}\text{ J} = 1.65\times10^{-10}\text{ eV} = 165\times10^{-12}\text{ eV}$$

$$= 165\text{ peV}$$

The uncertainty is 165 peV or more.

P41.50 Suppose the marble has mass 20 g. Suppose the wall of the box is 12 cm high and 2 mm thick. While it is inside the wall,

$$U = mgy = \left(0.02\text{ kg}\right)\left(9.8\text{ m/s}^2\right)\left(0.12\text{ m}\right) = 0.023\,5\text{ J}$$

and

$$E = K = \frac{1}{2}mu^2 = \frac{1}{2}\left(0.02\text{ kg}\right)\left(0.8\text{ m/s}\right)^2 = 0.006\,4\text{ J}$$

Then,

$$C = \frac{\sqrt{2m(U-E)}}{\hbar} = \frac{\sqrt{2\left(0.02\text{ kg}\right)\left(0.017\,1\text{ J}\right)}}{1.055\times10^{-34}\text{ J}\cdot\text{s}} = 2.5\times10^{32}\text{ m}^{-1}$$

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and the transmission coefficient is

$$e^{-2CL} = e^{-2\left(2.5\times10^{32}\right)\left(2\times10^{-3}\right)} = e^{-10\times10^{29}} = e^{-2.30\left(4.3\times10^{29}\right)}$$

$$= 10^{-4.3\times10^{29}} = \boxed{\sim 10^{-10^{30}}}$$

P41.51 (a) From Equation 41.14, the allowed energy levels are

$$E_n = \left(\frac{h^2}{8mL^2}\right)n^2 \qquad n = 1, 2, 3, \ldots$$

The energy of the absorbed photon is

$$E = \Delta E_n = E_3 - E_1 = \left(\frac{h^2}{8m_eL^2}\right)(3)^2 - \left(\frac{h^2}{8m_eL^2}\right)(1)^2 = 8\left(\frac{h^2}{8m_eL^2}\right)$$

We determine the length of the box from

$$\frac{hc}{\lambda} = \frac{h^2}{m_eL^2} \qquad \rightarrow \qquad \boxed{L = \left(\frac{h\lambda}{m_ec}\right)^{1/2}}$$

(b) The energy lost during the $n = 3$ to $n = 2$ transition is

$$E' = E_3 - E_2 = \left(\frac{h^2}{8m_eL^2}\right)(3)^2 - \left(\frac{h^2}{8m_eL^2}\right)(2)^2 = 5\left(\frac{h^2}{8m_eL^2}\right)$$

The wavelength of the emitted photon is then

$$\frac{hc}{\lambda'} = \frac{5h^2}{8m_eL^2} = \frac{5h^2}{8\cancel{m_e}}\left(\frac{\cancel{m_e}c}{h\lambda}\right) \quad \rightarrow \quad \boxed{\lambda' = \frac{8}{5}\lambda}$$

P41.52 $\left\langle x^2 \right\rangle = \int_{-\infty}^{\infty} x^2 |\psi|^2 \, dx$

For a one-dimensional box of width L, from Equation 41.18,

$$\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

$$\left\langle x^2 \right\rangle = \frac{2}{L}\int_0^L x^2 \sin^2\left(\frac{n\pi x}{L}\right)dx$$

With the substitution

$$y = \frac{n\pi x}{L} \;\rightarrow\; dy = \frac{n\pi}{L}dx$$

$$x = \frac{L}{n\pi}y \;\rightarrow\; dx = \frac{L}{n\pi}dy$$

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the integral becomes (from integral tables)

$$\langle x^2\rangle = \frac{2}{L}\left(\frac{L}{n\pi}\right)^3 \int_0^{n\pi} x^2 \sin^2 y\, dy$$

$$= \frac{2L^2}{(n\pi)^3}\left[\frac{y^3}{6} - \left(\frac{y^2}{4} - \frac{1}{8}\right)\sin 2y - \frac{y}{4}\cos 2y\right]\Bigg|_0^{n\pi}$$

$$= \frac{2L^2}{(n\pi)^3}\left[\frac{(n\pi)^3}{6} - \frac{n\pi}{4}\cos 2(n\pi)\right]$$

$$= \frac{2L^2}{(n\pi)^3}\left[\frac{(n\pi)^3}{6} - \frac{n\pi}{4}\right] = \frac{L^2}{3} - \frac{L^2}{2n^2\pi^2}$$

P41.53 (a) The requirements that $\frac{n\lambda}{2} = L$ and $p = \frac{h}{\lambda} = \frac{nh}{2L}$ are still valid. From the relativistic energy of the particle,

$$E = \sqrt{(pc)^2 + (mc^2)^2} \Rightarrow E_n = \sqrt{\left(\frac{nhc}{2L}\right)^2 + (mc^2)^2}$$

its kinetic energy is therefore

$$K_n = E_n - mc^2 = \boxed{\sqrt{\left(\frac{nhc}{2L}\right)^2 + (mc^2)^2} - mc^2}$$

(b) Taking $L = 1.00 \times 10^{-12}$ m, $m = 9.11 \times 10^{-31}$ kg, and $n = 1$, we find

$$K_n = \sqrt{\left(\frac{nhc}{2L}\right)^2 + (mc^2)^2} - mc^2$$

$$= \left\{\left[\frac{(1)(6.626\times10^{-34}\ \text{J}\cdot\text{s})(2.998\times10^{8}\ \text{m/s})}{2(1.00\times10^{-12}\ \text{m})}\right]^2 + \left[(9.11\times10^{-31}\ \text{kg})\left(2.998\times10^{8}\ \frac{\text{m}}{\text{s}}\right)^2\right]^2\right\}^{1/2} - (9.11\times10^{-31}\ \text{kg})\left(2.998\times10^{8}\ \frac{\text{m}}{\text{s}}\right)^2$$

$$= \boxed{4.68\times10^{-14}\ \text{J}}$$

(c) The particle's nonrelativistic energy is

$$E_1 = \frac{h^2}{8mL^2} = \frac{(6.626\times10^{-34}\ \text{J}\cdot\text{s})^2}{8(9.11\times10^{-31}\ \text{kg})(1.00\times10^{-12}\ \text{m})^2} = 6.02\times10^{-14}\ \text{J}$$

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Comparing this to K_1, we see that this value is too large by $\boxed{28.6\%}$.

P41.54 Looking at Figure 41.7, we see that wavelengths for a particle in a finite well are longer than those for a particle in an infinite well. Therefore, the energies of the allowed states should be lower for a finite well than for an infinite well. As a result, the photons from the source have too much energy to be absorbed or, equivalently, the photons have a frequency that is too high. In order to lower their apparent frequency using the Doppler shift, the source would have to move *away* from the particle in the finite square well, not *toward* it.

P41.55 (a) For a particle with wave function

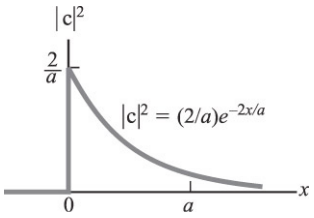
$$\psi(x) = \begin{cases} \sqrt{\frac{2}{a}} e^{-x/a} & \text{for } x > 0 \\ 0 & \text{for } x < 0 \end{cases}$$

The probability densities are

$$|\psi(x)|^2 = 0 \qquad \text{for } x < 0$$

and $$|\psi^2(x)| = \frac{2}{a} e^{-2x/a} \qquad \text{for } x > 0.$$

ANS. FIG. P41.55. shows a sketch of the probability density for this particle.



ANS. FIG. P41.55

(b) The probability is obtained from

$$\text{Prob}(x < 0) = \int_{-\infty}^{0} |\psi(x)|^2 \, dx = \int_{-\infty}^{0} (0) dx = \boxed{0}$$

(c) For the wave function to be normalized, we require

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = \int_{-\infty}^{0} |\psi|^2 \, dx + \int_{0}^{\infty} |\psi|^2 \, dx = 1$$

Performing the integration gives

$$\int_{-\infty}^{0} 0 dx + \int_{0}^{\infty} \left(\frac{2}{a}\right) e^{-2x/a} dx = 0 - e^{-2x/a}\Big|_0^{\infty} = -\left(e^{-\infty} - 1\right) = 1$$

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(d) The probability is obtained from

$$\text{Prob}(0 < x < a) = \int_0^a |\psi|^2 \, dx = \int_0^a \left(\frac{2}{a}\right) e^{-2x/a} \, dx = -e^{-2x/a}\Big|_0^a$$

$$= 1 - e^{-2} = \boxed{0.865}$$

P41.56 (a) Taking $L_x = L_y = L$, we see that the expression for E becomes

$$\boxed{E = \frac{h^2}{8m_e L^2}\left(n_x^2 + n_y^2\right)}$$

The general form of the wave function is

$$\psi \sim \sin\left(\frac{n_x \pi x}{L}\right)\sin\left(\frac{n_y \pi y}{L}\right)$$

For a normalizable wave function, neither n_x nor n_y can be zero, otherwise $\psi = 0$.

(b) The ground state corresponds to $\boxed{n_x = n_y = 1}$.

(c) The energy of the ground state is

$$E_{1,\,1} = \frac{h^2}{8m_e L^2}\left(1^2 + 1^2\right) = \boxed{\frac{h^2}{4m_e L^2}}$$

(d) For the first excited state, $\boxed{n_x = 1 \text{ and } n_y = 2, \text{ or } n_x = 2 \text{ and } n_y = 1}$.

(e) For the second excited state, $\boxed{n_x = 2 \text{ and } n_y = 2}$.

(f) The second excited state, corresponding to $n_x = 2$, $n_y = 2$, has an energy of

$$E_{2,\,2} = \frac{h^2}{8m_e L^2}\left(2^2 + 2^2\right) = \boxed{\frac{h^2}{m_e L^2}}$$

(g) The energy difference between the ground state and the second excited state is

$$\Delta E = E_{2,\,2} - E_{1,\,1} = \frac{h^2}{m_e L^2} - \frac{h^2}{4m_e L^2} = \boxed{\frac{3h^2}{4m_e L^2}}$$

(h) $$\Delta E = \frac{3h^2}{4m_e L^2} = \frac{hc}{\lambda} \quad \rightarrow \quad \lambda = \boxed{\frac{4m_e c L^2}{3h}}$$

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P41.57 (a) The expectation value is

$$\langle x\rangle_0 = \int_{-\infty}^{\infty} x\left(\frac{a}{\pi}\right)^{1/2} e^{-ax^2}\,dx = \boxed{0}$$

since the integrand is an odd function of x.

(b) The expectation value is

$$\langle x\rangle_1 = \int_{-\infty}^{\infty} x\left(\frac{4a^3}{\pi}\right)^{1/2} x^2 e^{-ax^2}\,dx = \boxed{0}$$

since the integrand is an odd function of x.

(c) The expectation value is

$$\langle x\rangle_{01} = \int_{-\infty}^{\infty} x\frac{1}{2}\left(\psi_0+\psi_1\right)^2 dx = \frac{1}{2}\langle x\rangle_0 + \frac{1}{2}\langle x\rangle_1 + \int_{-\infty}^{\infty} x\psi_0(x)\psi_1(x)\,dx$$

The first two terms are zero, from (a) and (b). Thus,

$$\langle x\rangle_{01} = \int_{-\infty}^{\infty} x\left(\frac{a}{\pi}\right)^{1/4} e^{-ax^2/2}\left(\frac{4a^3}{\pi}\right)^{1/4} xe^{-ax^2/2}\,dx = 2\left(\frac{2a^2}{\pi}\right)^{1/2}\int_0^{\infty} x^2 e^{-ax^2}\,dx$$

$$= 2\left(\frac{2a^2}{\pi}\right)^{1/2}\frac{1}{4}\left(\frac{\pi}{a^3}\right)^{1/2}$$

$$= \boxed{\frac{1}{\sqrt{2a}}}$$

Where we have used Table B.6 in the Appendix to evaluate the integral.

P41.58 With one slit open,

$$P_1 = \left|\psi_1\right|^2 \quad \text{or} \quad P_2 = \left|\psi_2\right|^2$$

With both slits open,

$$P = \left|\psi_1+\psi_2\right|^2$$

At a maximum, the wave functions are in phase

$$P_{\max} = \left(\left|\psi_1\right|+\left|\psi_2\right|\right)^2$$

At a minimum, the wave functions are out of phase,

$$P_{\min} = \left(\left|\psi_1\right|-\left|\psi_2\right|\right)^2$$

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Now,

$$\frac{P_1}{P_2} = \frac{|\psi_1|^2}{|\psi_2|^2} = 25.0\,,$$

so $$\frac{|\psi_1|}{|\psi_2|} = 5.00$$

and $$\frac{P_{\max}}{P_{\min}} = \frac{\left(|\psi_1| + |\psi_2|\right)^2}{\left(|\psi_1| - |\psi_2|\right)^2} = \frac{\left(5.00|\psi_2| + |\psi_2|\right)^2}{\left(5.00|\psi_2| - |\psi_2|\right)^2} = \frac{(6.00)^2}{(4.00)^2} = \frac{36.0}{16.0} = \boxed{2.25}$$

Challenge Problems

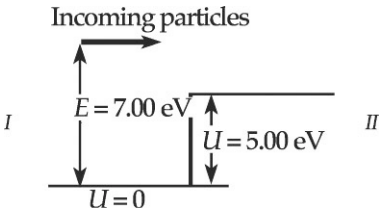
P41.59 (a) The claim is that Schrödinger's equation

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{2m}{\hbar^2}(E - U)\psi$$

has the solutions

$$\psi_1 = Ae^{ik_1x} + Be^{-ik_1x} \quad \text{[region } I\text{]}$$

$$\psi_2 = Ce^{ik_2x} \quad \text{[region } II\text{]}$$



ANS. FIG. P41.59(a)

Check that the solution for region *I* satisfies Schrödinger's equation:

$$\frac{\partial^2\psi_1}{\partial x^2} = -\frac{2m}{\hbar^2}E\psi_1$$

$$\frac{\partial^2}{\partial x^2}\left(Ae^{ik_1x}\right) + \frac{\partial^2}{\partial x^2}\left(Be^{-ik_1x}\right) = -\frac{2m}{\hbar^2}E\left(Ae^{ik_1x} + Be^{-ik_1x}\right)$$

$$-k_1^2\left(Ae^{ik_1x}\right) - k_1^2\left(Be^{-ik_1x}\right) = -\frac{2m}{\hbar^2}E\left(Ae^{ik_1x} + Be^{-ik_1x}\right)$$

$$-k_1^2\left(Ae^{ik_1x} + Be^{-ik_1x}\right) = -\frac{2m}{\hbar^2}E\left(Ae^{ik_1x} + Be^{-ik_1x}\right)$$

The last line is true if $k_1^2 = \frac{2m}{\hbar^2}E$, which it is because

$$E = \frac{p^2}{2m} = \frac{\left(\hbar k_1\right)^2}{2m} \;\rightarrow\; k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Therefore, the equation is satisfied in region *I*.

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Check that the solution for region *II* satisfies Schrödinger's equation:

$$\frac{\partial^2\psi_2}{\partial x^2} = -\frac{2m}{\hbar^2}(E-U)\psi_2$$

$$\frac{\partial^2}{\partial x^2}\left(Ce^{ik_2x}\right) = -\frac{2m}{\hbar^2}(E-U)\left(Ce^{ik_2x}\right)$$

$$-k_2^2\left(Ce^{ik_2x}\right) = -\frac{2m}{\hbar^2}(E-U)\left(Ce^{ik_2x}\right)$$

The last line is true if $k_2^2 = \frac{2m}{\hbar^2}(E-U)$, which it is because

$$E = \frac{p^2}{2m} + U = \frac{\left(\hbar k_2\right)^2}{2m} \rightarrow k_2 = \frac{\sqrt{2m(E-U)}}{\hbar}$$

Therefore, the equation is satisfied in region *II*. We apply boundary conditions. Matching functions and derivatives at $x = 0$, we find that

$$\left(\psi_1\right)_0 = \left(\psi_2\right)_0 \qquad \text{gives} \quad A + B = C,$$

and $\left(\frac{d\psi_1}{dx}\right)_0 = \left(\frac{d\psi_2}{dx}\right)_0$ gives $k_1(A-B) = k_2C.$

Then $B = \frac{1-k_2/k_1}{1+k_2/k_1}A$ and $C = \frac{2}{1+k_2/k_1}A$.

Incident wave Ae^{ikx} reflects Be^{-ikx}, with probability

$$R = \frac{B^2}{A^2} = \frac{\left(1-k_2/k_1\right)^2}{\left(1+k_2/k_1\right)^2} = \boxed{\frac{\left(k_1-k_2\right)^2}{\left(k_1+k_2\right)^2}}$$

(b) With $E = 7.00$ eV and $U = 5.00$ eV:

$$\frac{k_2}{k_1} = \sqrt{\frac{E-U}{E}} = \sqrt{\frac{2.00\text{ eV}}{7.00\text{ eV}}} = 0.535$$

The reflection probability is $R = \frac{(1-0.535)^2}{(1+0.535)^2} = \boxed{0.092\ 0}$.

(c) The probability of transmission is $T = 1 - R = \boxed{0.908}$.

P41.60 (a) The potential energy of the system is given by

$$U = \frac{e^2}{4\pi\in_0 d}\left[\left(-1+\frac{1}{2}-\frac{1}{3}\right)+\left(-1+\frac{1}{2}\right)+(-1)\right] = \frac{(-7/3)e^2}{4\pi\in_0 d}$$

$$= \boxed{-\frac{7k_ee^2}{3d}}$$

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(b) There are two electrons, each with minimum energy E_1. From Equation 41.14, the total energy is

$$K = 2E_1 = \frac{2h^2}{8m_e(3d)^2} = \boxed{\frac{h^2}{36m_e d^2}}$$

(c) The total energy of the system is

$$E = K + U = \frac{h^2}{36m_e d^2} - \frac{7k_e e^2}{3d}$$

For a minimum, we require $\frac{dE}{d(d)} = 0$. Differentiating,

$$\frac{dE}{d(d)} = 0$$

$$\frac{d}{d(d)}\left(\frac{h^2}{36m_e d^2} - \frac{7k_e e^2}{3d}\right) = 0$$

$$(-2)\frac{h^2}{36m_e d^3} - (-1)\frac{7k_e e^2}{3d^2} = 0$$

$$\frac{h^2}{18m_e d^3} = \frac{7k_e e^2}{3d^2}$$

$$d = \frac{3h^2}{7(18m_e)k_e e} = \frac{h^2}{42m_e k_e e^2}$$

Substituting numerical values,

$$d = \frac{\left(6.626\times10^{-34}\ \text{J}\cdot\text{s}\right)^2}{(42)\left(9.11\times10^{-31}\ \text{kg}\right)\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(1.60\times10^{-19}\ \text{C}\right)^2}$$

$$= 4.99\times10^{-11}\ \text{m} = \boxed{49.9\ \text{pm}}$$

(d) The lithium spacing is d and the number of atoms N in volume V is related by $Nd^3 = V$, and the density is $\frac{Nm}{V}$, where m is the mass of one atom. We have:

$$\text{density} = \frac{Nm}{V} = \frac{Nm}{Nd^3} = \frac{m}{d^3}$$

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From which we obtain

$$d = \left(\frac{m}{\text{density}}\right)^{1/3} = \left[\frac{6.94\ \text{g}}{\text{mol}}\left(\frac{1\ \text{mol}}{6.022\times10^{23}\ \text{atoms}}\right)\Big/\ 0.530\ \frac{\text{g}}{\text{cm}^3}\right]^{1/3}$$

$$= 2.79\times10^{-8}\ \text{cm} = 2.79\times10^{-10}\ \text{mm} = 279\ \text{pm}$$

The lithium interatomic spacing of 280 pm is 5.59 times larger. Therefore, it is of the same order of magnitude as the interatomic spacing $2d$ here.

P41.61 The wave functions and probability densities are the same as those shown in Active Figure 41.4 of the textbook. From Equation 41.13, the wave functions are

$$\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right) \text{ where } n = 1, 2, 3\ldots$$

(a) For $n = 1$,

$$P_1 = \int_{0.150\ \text{nm}}^{0.350\ \text{nm}} |\psi_1|^2\,dx = \left(\frac{2}{1.00\ \text{nm}}\right)\int_{0.150}^{0.350}\sin^2\left(\frac{\pi x}{1.00\ \text{nm}}\right)dx$$

$$= (2.00/\text{nm})\left[\frac{x}{2} - \frac{1.00\ \text{nm}}{4\pi}\sin\left(\frac{2\pi x}{1.00\ \text{nm}}\right)\right]_{0.150\ \text{nm}}^{0.350\ \text{nm}}$$

In the above result we used

$$\int \sin^2(ax)\,dx = \left(\frac{x}{2}\right) - \left(\frac{1}{4a}\right)\sin(2ax)$$

Therefore,

$$P_1 = (1.00/\text{nm})\left[x - \frac{1.00\ \text{nm}}{2\pi}\sin\left(\frac{2\pi x}{1.00\ \text{nm}}\right)\right]_{0.150\ \text{nm}}^{0.350\ \text{nm}}$$

$$P_1 = (1.00/\text{nm})\left\{0.350\ \text{nm} - 0.150\ \text{nm} - \frac{1.00\ \text{nm}}{2\pi}\left[\sin(0.700\pi) - \sin(0.300\pi)\right]\right\}$$

$$= \boxed{0.200}$$

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(b) $$P_2 = \frac{2}{1.00}\int_{0.150}^{0.350} \sin^2\left(\frac{2\pi x}{1.00}\right)dx = 2.00\left[\frac{x}{2} - \frac{1.00}{8\pi}\sin\left(\frac{4\pi x}{1.00}\right)\right]_{0.150}^{0.350}$$

$$P_2 = 1.00\left[x - \frac{1.00}{4\pi}\sin\left(\frac{4\pi x}{1.00}\right)\right]_{0.150}^{0.350}$$

$$= 1.00\left\{(0.350 - 0.150) - \frac{1.00}{4\pi}\left[\sin(1.40\pi) - \sin(0.600\pi)\right]\right\}$$

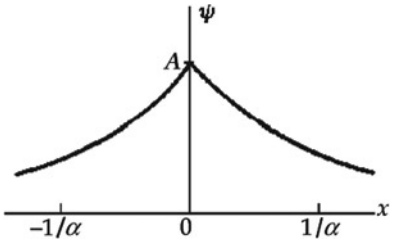
$$= \boxed{0.351}$$

Using $E_n = \dfrac{n^2h^2}{8mL^2}$, we find that

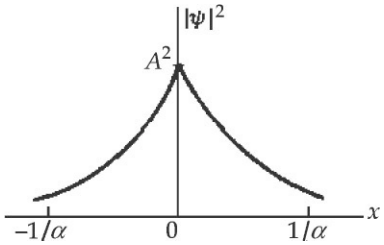
(c) $E_1 = \boxed{0.377 \text{ eV}}$ and

(d) $E_2 = \boxed{1.51 \text{ eV}}$

P41.62 (a) and (b) The Wave functions are shown in ANS. FIG. P41.62(a) and ANS. FIG. P41.62(b).



ANS. FIG. P41.62(a)



ANS. FIG. P41.62(b)

(c) ψ is continuous and $\psi \to 0$ as $x \to \pm\infty$. The function can be normalized. It describes a particle bound near $x = 0$.

(d) Since ψ is symmetric,

$$\int_{-\infty}^{\infty} |\psi|^2\, dx = 2\int_{0}^{\infty} |\psi|^2\, dx = 1$$

or $$2A^2\int_0^{\infty} e^{-2\alpha x}\, dx = \left(\frac{2A^2}{-2\alpha}\right)\left(e^{-\infty} - e^{0}\right) = 1.$$

This gives $\boxed{A = \sqrt{\alpha}}$.

(e) The probability of finding the particle between $-1/2\alpha$ and $+1/2\alpha$ is

$$P_{(-1/2\alpha)\to(1/2\alpha)} = 2\left(\sqrt{a}\right)^2\int_{x=0}^{1/2\alpha} e^{-2\alpha x}\, dx = \left(\frac{2\alpha}{-2\alpha}\right)\left(e^{-2\alpha/2\alpha} - 1\right)$$

$$= \left(1 - e^{-1}\right) = \boxed{0.632}$$

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P41.63 (a) Recall from Section 41.7 that the potential energy of a harmonic oscillator is $\frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2$. We can find the energy of the oscillator E by substituting the wave function into the Schrödinger equation.

$$\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi \quad \rightarrow \quad \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

From $\psi = Bxe^{-(m\omega/2\hbar)x^2}$, we have

$$\frac{d\psi}{dx} = Be^{-(m\omega/2\hbar)x^2} + Bx\left(-\frac{m\omega}{2\hbar}\right)2xe^{-(m\omega/2\hbar)x^2}$$

$$= Be^{-(m\omega/2\hbar)x^2} - B\left(\frac{m\omega}{\hbar}\right)x^2 e^{-(m\omega/2\hbar)x^2}$$

$$\frac{d^2\psi}{dx^2} = Bx\left(-\frac{m\omega}{\hbar}\right)xe^{-(m\omega/2\hbar)x^2} - B\left(\frac{m\omega}{\hbar}\right)2xe^{-(m\omega/2\hbar)x^2}$$

$$- B\left(\frac{m\omega}{\hbar}\right)x^2\left(-\frac{m\omega}{\hbar}\right)xe^{-(m\omega/2\hbar)x^2}$$

$$\frac{d^2\psi}{dx^2} = -3B\left(\frac{m\omega}{\hbar}\right)xe^{-(m\omega/2\hbar)x^2} + B\left(\frac{m\omega}{\hbar}\right)^2 x^3 e^{-(m\omega/2\hbar)x^2}$$

Substituting the above into the Schrödinger equation, we have

$$\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

$$\frac{-\hbar^2}{2m}\left[-3B\left(\frac{m\omega}{\hbar}\right)xe^{-(m\omega/2\hbar)x^2} + B\left(\frac{m\omega}{\hbar}\right)^2 x^3 e^{-(m\omega/2\hbar)x^2}\right]$$

$$+ \frac{1}{2}m\omega^2 x^2\left[Bxe^{-(m\omega/2\hbar)x^2}\right]$$

$$= E\left[Bxe^{-(m\omega/2\hbar)x^2}\right]$$

$$\left(\frac{3\hbar\omega}{2}\right)\left[Bxe^{-(m\omega/2\hbar)x^2}\right] + \cancel{\left(-\frac{1}{2}m\omega^2 x^2\right)\left[Bxe^{-(m\omega/2\hbar)x^2}\right]}$$

$$+ \cancel{\left(\frac{1}{2}m\omega^2 x^2\right)\left[Bxe^{-(m\omega/2\hbar)x^2}\right]}$$

$$= E\left[Bxe^{-(m\omega/2\hbar)x^2}\right]$$

$$\left(\frac{3\hbar\omega}{2}\right)\left(Bxe^{-(m\omega/2\hbar)x^2}\right) = E\left(Bxe^{-(m\omega/2\hbar)x^2}\right)$$

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The last line is true if $\boxed{E = \frac{3\hbar\omega}{2}}$.

(b) We never find the particle at $\boxed{x = 0}$ because $\psi = 0$ there.

(c) ψ is maximized if

$$\frac{d\psi}{dx} = Be^{-(m\omega/2\hbar)x^2} - B\left(\frac{m\omega}{\hbar}\right)x^2 e^{-(m\omega/2\hbar)x^2} = 0$$

$$1 - \left(\frac{m\omega}{\hbar}\right)x^2 = 0$$

which is true at $\boxed{x = \pm\sqrt{\frac{\hbar}{m\omega}}}$.

(d) We require $\int_{-\infty}^{\infty} |\psi|^2 \, dx = 1$:

$$1 = \int_{-\infty}^{\infty} B^2 x^2 e^{-(m\omega/\hbar)x^2} dx = 2B^2 \int x^2 e^{-(m\omega/\hbar)x^2} dx$$

$$= 2B^2 \frac{1}{4}\sqrt{\frac{\pi}{(m\omega/\hbar)^3}} = B^2 \frac{\pi^{1/2}}{2}\left(\frac{\hbar}{m\omega}\right)^{3/2}$$

Then,

$$B = \frac{2^{1/2}}{\pi^{1/4}}\left(\frac{m\omega}{\hbar}\right)^{3/4} = \boxed{\left(\frac{4m^3\omega^3}{\pi\hbar^3}\right)^{1/4}}$$

(e) At $x = 2(\hbar/m\omega)^{1/2}$, the potential energy is

$$\frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2\left(\frac{4\hbar}{m\omega}\right) = 2\hbar\omega$$

This is larger than the total energy $\frac{3\hbar\omega}{2}$, so there is $\boxed{\text{zero}}$ classical probability of finding the particle here.

(f) The actual probability is given by

$$P = |\psi|^2 \, dx = \left(Bxe^{-(m\omega/2\hbar)x^2}\right)^2 \delta$$

$$P = \delta B^2 x^2 e^{-(m\omega/\hbar)x^2} = \delta\left(\frac{4m^3\omega^3}{\pi\hbar^3}\right)^{1/2}\left(\frac{4\hbar}{m\omega}\right)e^{-(m\omega/\hbar)x^2}$$

$$= \delta\frac{2}{\pi^{1/2}}\left(\frac{m^{3/2}\omega^{3/2}}{\hbar^{3/2}}\right)\left(\frac{4\hbar}{m\omega}\right)e^{-(m\omega/\hbar)4(\hbar/m\omega)} = \boxed{8\delta\left(\frac{m\omega}{\hbar\pi}\right)^{1/2}e^{-4}}$$

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P41.64 (a) To find the normalization constant, we note that $\int_0^L |\psi|^2\,dx = 1$, or

$$A^2\int_0^L\left[\sin^2\left(\frac{\pi x}{L}\right)+16\sin^2\left(\frac{2\pi x}{L}\right)+8\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{2\pi x}{L}\right)\right]dx = 1$$

Noting that

$$\int_0^L \sin^2\left(\frac{\pi x}{L}\right)dx = \int_0^L \frac{1-\cos\left[2\left(\pi x/L\right)\right]}{2}dx$$

$$= \left[\frac{x}{2}-\frac{L}{\pi}\frac{\sin\left(2\pi x/L\right)}{2}\right]_0^L = \frac{L}{2}$$

the integral becomes

$$\int_0^L |\psi|^2\,dx = A^2\left[\left(\frac{L}{2}\right)+16\left(\frac{L}{2}\right)+8\int_0^L \sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{2\pi x}{L}\right)dx\right]$$

$$1 = A^2\left\{\left(\frac{L}{2}\right)+16\left(\frac{L}{2}\right)+8\int_0^L \sin\left(\frac{\pi x}{L}\right)\left[2\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)\right]dx\right\}$$

$$1 = A^2\left[\frac{17L}{2}+16\int_0^L \sin^2\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx\right]$$

$$1 = A^2\left[\frac{17L}{2}+\frac{16L}{3\pi}\sin^3\left(\frac{\pi x}{L}\right)\Big|_{x=0}^{x=L}\right] = A^2\left(\frac{17L}{2}\right)$$

$$\rightarrow \boxed{A = \sqrt{\frac{2}{17L}}}$$

(b) To determine the relationship between A and B, we note that $\int_{-a}^{a} |\psi|^2\,dx = 1$. Therefore,

$$\int_{-a}^{a}\left[|A|^2\cos^2\left(\frac{\pi x}{2a}\right)+|B|^2\sin^2\left(\frac{\pi x}{a}\right)+2|A||B|\cos\left(\frac{\pi x}{2a}\right)\sin\left(\frac{\pi x}{a}\right)\right]dx = 1$$

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Noting that

$$\int_{-a}^{a} \sin^2\left(\frac{\pi x}{2a}\right) dx = \int_{-a}^{a} \frac{1-\cos\left[2\left(\pi x/2a\right)\right]}{2} dx$$

$$= \left[\frac{x}{2} + \frac{2L}{\pi}\frac{\sin\left(\pi x/a\right)}{2}\right]\Bigg|_{-a}^{a} = a$$

and

$$\int_{-a}^{a} \cos^2\left(\frac{\pi x}{2a}\right) dx = \int_{-a}^{a} \frac{1+\cos\left[2\left(\pi x/2a\right)\right]}{2} dx$$

$$= \left[\frac{x}{2} + \frac{2L}{\pi}\frac{\sin\left(\pi x/a\right)}{2}\right]\Bigg|_{-a}^{a} = a$$

the integral becomes

$$|A|^2 a + |B|^2 a + \int_{-a}^{a}\left[2|A||B|\cos\left(\frac{\pi x}{2a}\right)\sin\left(\frac{\pi x}{a}\right)\right] dx = 1$$

The third term is:

$$2|A||B|\int_{-a}^{a}\cos\left(\frac{\pi x}{2a}\right)\left[2\sin\left(\frac{\pi x}{2a}\right)\cos\left(\frac{\pi x}{2a}\right)\right] dx$$

$$= 4|A||B|\int_{-a}^{a}\cos^2\left(\frac{\pi x}{2a}\right)\sin\left(\frac{\pi x}{2a}\right) dx$$

$$= \frac{8a|A||B|}{3\pi}\cos^3\left(\frac{\pi x}{2a}\right)\Bigg|_{-a}^{a} = 0$$

so the whole integral is

$$a\left(|A|^2 + |B|^2\right) = 1\text{, giving } \boxed{|A|^2 + |B|^2 = \frac{1}{a}}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P41.2 (a) See ANS. FIG. P41.2; (b) $\frac{1}{\sqrt{a}}$; (c) 0.865

P41.4 $\frac{1}{2}$

P41.6 (a) $n \approx 4$; (b) 6.03 eV

P41.8 The photon does not have the smallest possible energy to cause the transition between states $n = 1$ to $n = 2$.

P41.10 $\sqrt{\frac{3h\lambda}{8m_e c}}$

P41.12 (a) 5.13×10^{-3} eV; (b) 9.41 eV; (c) The electron has a much higher energy because it is much less massive.

P41.14 (a) 2.00×10^{-9} J; (b) 1.66×10^{-28} m; (c) No. The length of the box would have to be much smaller than the size of a nucleus ($\sim 10^{-14}$ m) to confine the particle.

P41.16 (a) $\frac{\hbar}{2L}$; (b) $\hbar^2/8mL^2$; (c) This estimate is too low by $4\pi^2 \approx 40$ times, but it correctly displays the pattern of dependence of the energy on the mass and on the length of the well.

P41.18 See P41.18 for full explanation.

P41.20 (a) $x = L/4$, $L/2$, and $3L/4$; (b) We look for $\sin(3\pi x/L)$ taking on its extreme values 1 and –1 so that the squared wave function is as large as it can be. The result can also be found by studying Figure 41.4b.

P41.22 (a) $\frac{\ell}{L} - \frac{1}{2\pi}\sin\left(\frac{2\pi\ell}{L}\right)$; (b) See ANS FIG P41.22(b); (c) The wave function is zero for $x < 0$ and for $x > L$. The probability at $\ell = 0$ must be zero because the particle is never found at $x < 0$ or exactly at $x = 0$. The probability at $\ell = L$ must be 1 for normalization: the particle is always found somewhere in the range $0 < x < L$; (d) $0.585L$

P41.24 See P41.24 for complete solution.

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P41.26 (a) $n = 1$: $\psi_1(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{\pi x}{L}\right)$; $P_1(x) = |\psi_1(x)|^2 = \frac{2}{L}\cos^2\left(\frac{\pi x}{L}\right)$,

$n = 2$: $\psi_2(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right)$; $P_2(x) = |\psi_2(x)|^2 = \frac{2}{L}\sin^2\left(\frac{2\pi x}{L}\right)$,

$n = 3$: $\psi_3(x) = \sqrt{\frac{2}{L}}\cos\left(\frac{3\pi x}{L}\right)$; $P_3(x) = |\psi_3(x)|^2 = \frac{2}{L}\cos^2\left(\frac{3\pi x}{L}\right)$;

(b) See ANS FIG. P41.26(b).

P41.28 (a) $\frac{\hbar^2}{L^2 m}$; (b) $\sqrt{\frac{15}{16L}}$; (c) 0.580

P41.30 (a) See ANS. FIG. P41.30(a); (b) $2L$

P41.32 (a) 1.03×10^{-3}; (b) 1.91 nm

P41.34 1.35

P41.36 (a) See P41.36(a) for full explanation; (b) $b = \frac{m\omega}{2\hbar}$ and $\frac{3}{2}\hbar\omega$;

(c) first excited state

P41.38 $2\pi c\sqrt{\frac{m}{k}}$

P41.40 (a) See P41.40(a) for full explanation; (b) See P41.40(b) for full explanation; (c) $f = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$

P41.42 See P41.42 for full explanation.

P41.44 (a–b) See P41.44(a) and (b) for full explanations.

P41.46 See P41.46 for full explanation.

P41.48 (a) See P41.48(a) for full proof; (b) For 2 to 1, λ = 1.38 μm, infrared; For 3 to 2, λ = 827 nm, infrared; For 4 to 1, λ = 275 nm, ultraviolet; For 4 to 2, λ = 344 nm, near ultraviolet; For 4 to 3, λ = 590 nm, yellow-orange visible.

P41.50 $\sim 10^{-10^{30}}$

P41.52 See P41.52 for full explanation.

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P41.54 Looking at Figure 41.7, we see that wavelengths for a particle in a finite well are longer than those for a particle in an infinite well. Therefore, the energies of the allowed states should be lower for a finite well than for an infinite well. As a result, the photons from the source have too much energy to be absorbed or, equivalently, the photons have a frequency that is too high. In order to lower their apparent frequency using the Doppler shift, the source would have to move *away* from the particle in the finite square well, not *toward* it.

P41.56 (a) $E = \dfrac{h^2}{8m_e L^2}\left(n_x^2 + n_y^2\right)$; (b) $n_x = n_y = 1$; (c) $\dfrac{h^2}{4m_e L^2}$; (d) $n_x = 1$ and $n_y = 2$, or $n_x = 2$ and $n_y = 1$; (e) $n_x = 2$ and $n_y = 2$; (f) $\dfrac{h^2}{m_e L^2}$; (g) $\dfrac{3h^2}{4m_e L^2}$; (h) $\dfrac{4m_e cL^2}{3h}$

P41.58 2.25

P41.60 (a) $-\dfrac{7k_e e^2}{3d}$; (b) $\dfrac{h^2}{36m_e d^2}$; (c) 49.9 pm; (d) The lithium interatomic spacing of 280 pm is 5.59 times larger. Therefore, it is of the same order of magnitude as the interatomic spacing $2d$ here.

P41.62 (a) See ANS. FIG. P41.62(a); (b) See ANS. FIG. P41.62(b); (c) ψ is continuous and $\psi \to 0$ as $x \to \pm\infty$. The function can be normalized. It describes a particle bound near $x = 0$; (d) $A = \sqrt{\alpha}$; (e) 0.632

P41.64 (a) $A = \sqrt{\dfrac{2}{17L}}$; (b) $|A|^2 + |B|^2 = \dfrac{1}{a}$

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